

University of Asia Pacific
Department of Computer Science and Engineering

Design, Implementation and Verification of a 4-Bit Adder/Subtractor ALU in CMOS

CMOS Implementation of a 4-bit Datapath, Track A: 4-bit ALU

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1. Project Title

Design, Implementation and Verification of a 4-Bit Adder/Subtractor ALU in CMOS

Track A (4-bit ALU) of the lab project "CMOS Implementation of a 4-bit Datapath", CSE 450, Design and Testing of VLSI Lab. The ALU performs two opcode-selected operations, addition and subtraction, and is designed at transistor level in static CMOS in Cadence Virtuoso using the gpdk045 (45 nm) process design kit.

2. Abstract

This report presents the design, implementation and functional verification of a 4-bit combinational arithmetic logic unit (ALU) that performs addition and subtraction, built at transistor level in static CMOS with Cadence Virtuoso in the gpdk045 (45 nm) technology. The unit accepts two 4-bit operands A and B and a 1-bit operation select OP, and produces a 4-bit result R and a single flag OVF that signals carry-out during addition and borrow during subtraction. Subtraction is realised by reusing the adder: B is conditionally inverted by an XOR gate controlled by OP, and OP is also supplied as the first carry-in, which implements two's complement arithmetic with no additional adder hardware.

The design is hierarchical. Inverter, NAND2, NOR2 and XOR2 cells form a full adder (58 transistors); a full adder plus one XOR2 forms a one-bit slice (72 transistors); four slices and an output XOR2 form the ALU (302 transistors). Using Spectre transient simulation in ADE with a 1.8 V supply and a 10 fF load on every output, the one-bit slice was verified over all 16 input combinations and the complete ALU over 13 directed test cases that include zero operands, all-ones operands, carry-out, borrow and equal-operand subtraction. All 16 slice combinations and all 13 ALU cases agree with the expected values, and the outputs settle to full logic levels (1.8 V high, 141 nV low at OVF). The worst-case carry-path propagation delay is 275.7 ps and the average power over the 13-case run is about 10.25 µW, both at a 1.8 V supply.

3. Introduction

3.1 Background and Motivation

A datapath is the part of a digital processor that holds, moves and transforms data. It is built from storage (registers or memory), selection logic (multiplexers) and an arithmetic logic unit, and it is steered by control signals from a control unit. Datapath design matters because the datapath sets most of a processor's speed, area and power; in particular, the carry path through the ALU is usually the longest combinational path and limits the clock frequency.

The arithmetic logic unit is the computational core of every processor, and the adder is its most fundamental building block. Implementing an adder and subtractor at transistor level is a standard exercise in digital VLSI design [1], [2] because it exposes the full design flow in a compact form: choosing a logic structure, composing it into reusable cells, capturing it as schematics and symbols, and proving its behaviour with transient simulation under realistic loading. Of the four datapath components offered in this lab, this project implements Track A, the 4-bit ALU, with the two operations required for this phase: addition and subtraction.

3.2 Scope and Organisation

The work covers schematic-level design and transient simulation with a 10 fF load on every output. Physical layout and parasitic extraction are outside the scope of the results reported here. The propagation delay and the average power of the schematic are reported in Section 9.

Section 4 states the objectives and Section 5 the system specification. Section 6 explains the arithmetic and the architecture. Section 7 describes the schematics, symbols and their implementation in Cadence Virtuoso, including transistor sizes. Section 8 describes the simulation and verification, Section 9 the performance analysis, and Section 10 concludes. Appendix A gives the stimulus definition of the ALU testbench, Appendix B lists the deviations from the project proposal, and Appendix C documents the errors found and corrected during the build. All screenshots come from the team's own Cadence session. The block diagrams (Figure 1, Figure 2, Figure 3, Figure 4) are drawings, and Figure 22 is a composite of two simulator screenshots with added labels (see Section 8.6).

4. Objectives

The objective of this project is to design a 4-bit ALU in static CMOS that computes R = A + B when OP = 0 and R = A − B when OP = 1, and that reports through OVF whenever the result is not a valid unsigned value, that is, on carry-out in addition and on borrow in subtraction. The specific objectives are:

1. Derive the Boolean equations of the result bits R3 to R0, the internal carries C1 to C4 and the flag OVF for both operations, before any transistor is drawn.
2. Decompose the equations into static CMOS cells (inverter, NAND2, NOR2, XOR2) and document the transistor sizes, supply voltage, load capacitance and input transition assumptions.
3. Design and verify a transistor-level full adder and a one-bit add/subtract slice (XOR-controlled inversion of B followed by a full adder).
4. Chain four slices into a ripple-carry adder/subtractor and add the OVF logic.
5. Create a symbol for each level of the hierarchy and separate testbench cells for the slice and the 4-bit ALU.
6. Apply at least ten test cases (13 are used for the ALU and 16 for the slice) in ADE transient analysis and compare the simulated outputs with expected values calculated beforehand.
7. Evaluate the design: transistor count as an area estimate, output voltage levels, and, as planned in the proposal, propagation delay and power (see Section 9 for what was and was not achieved).

5. System Specification

5.1 Input and Output Signals

The ALU has nine inputs and five outputs, as required by the Track A signal table of the lab guideline. The inputs are OP (operation select), A0 to A3 (operand A) and B0 to B3 (operand B). The outputs are R0 to R3 (result) and OVF (flag). Two supply pins, vdd and gnd, complete the interface. The circuit is purely combinational: it has no clock and no state, and its outputs settle a short time after any input change (Figure 1). Table 1 maps each guideline field to the pins of the Cadence symbol.

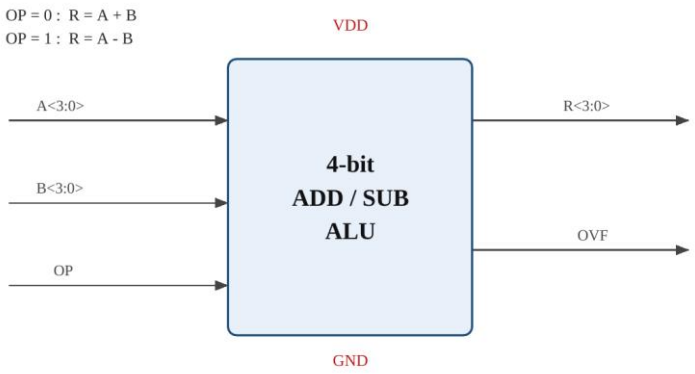


Figure 1. Top-level view of the ALU. Pins: A0–A3, B0–B3 and OP in; R0–R3 and OVF out; vdd and gnd supply.

Table 1. Interface specification and mapping of the guideline signal fields to the symbol pins.

Guideline field	Symbol pin	Width	Dir.	Description
input[0]	OP	1	In	Operation select: 0 = ADD, 1 = SUB
input[1:4]	A0–A3	4	In	First operand (A3 is the MSB)
input[5:8]	B0–B3	4	In	Second operand (B3 is the MSB)
output[0:3]	R0–R3	4	Out	ALU result (R3 is the MSB)
output[4]	OVF	1	Out	Overflow/error bit: carry-out for ADD, borrow for SUB
–	vdd, gnd	1 each	Power	Supply rails, 1.8 V and 0 V

5.2 Operations

Table 2. Operation select and meaning of the flag.

OP	Operation	Implementation	OVF = 1 indicates
0	R = A + B	B passes unchanged, carry-in = 0	Carry-out: A + B exceeds 15
1	R = A − B	B is inverted, carry-in = 1 (two's complement, A + B' + 1)	Borrow: A is smaller than B

5.3 Technology and Operating Conditions

Table 3 lists the technology parameters and the conditions under which every simulation in this report was run. The values come from the Cadence netlists and schematics of the design.

Table 3. Technology parameters and operating conditions.

Parameter	Value
Technology	gpdk045 (Cadence 45 nm generic process design kit), library gpdk045
Devices	NMOS nmos1v (model g45n1svt) and PMOS pmos1v (model g45p1svt)
Transistor size	W = 120 nm, L = 45 nm, one finger, multiplier 1, for every transistor (see Section 7.2)
Supply	VDD = 1.8 V, GND = 0 V (vdc source: V9 in the ALU testbench, V4 in the slice testbench)
Bulk connections	NMOS bulk to ground, PMOS bulk to the positive supply, inside every cell
Load capacitance	10 fF from each output (R0–R3 and OVF in the ALU; R and Cout in the slice) to ground
Input transitions	ALU testbench: vpwl sources, 0.1 ns edges. Slice testbench: vpulse sources with tr = 100 ps
Temperature	27 °C

Parameter	Value
Model file	gpdk045.scs, section mc (as included in the generated Spectre netlist); Maestro corner nom
Simulator and analysis	Cadence Spectre through ADE, transient analysis: 160 ns for the slice, 150 ns for the ALU

5.4 Supply Voltage

All simulations use a 1.8 V supply. This value was inherited from the project proposal, which was written for a 180 nm technology; the design was then built in gpdk045, whose devices are normally operated at about 1.0 V. The 1.8 V supply was kept for the following reason: every waveform, table and verdict in this report was produced at 1.8 V, and the functional behaviour of a static CMOS gate network does not depend on the particular supply value as long as the devices conduct. The ALU was observed to produce full logic levels at this supply.

The limitation must be stated clearly. Operation at the kit's nominal supply was not simulated, so no claim is made about correct operation at about 1.0 V. A supply of 1.8 V is well above the nominal value for these devices, so any delay or power figure obtained at 1.8 V would not represent nominal operation. For the same reason, the delay and power reported in Section 9 are valid only at 1.8 V, and a re-run at the nominal supply is listed as future work.

5.5 Performance Requirements

Neither the lab guideline nor the proposal sets numerical delay, power or area targets. The requirements applied in this work are therefore the following.

- Functional: the ALU shall produce the expected R and OVF for every directed test case, in both ADD and SUB modes, including the boundary cases. At least ten cases are required by the guideline; 13 are used.
- Output levels: R and OVF shall reach full logic levels (near VDD for logic 1 and near 0 V for logic 0) with the 10 fF load.
- Settling: outputs shall be settled before the end of each 10 ns test window, which is where the results are read.
- Area: the transistor count shall be reported as the area estimate, since no layout is made.
- Delay and power: the proposal planned to measure the worst-case carry-path delay and the average supply power. Section 9 reports the measured values.

5.6 Design Constraints

- Static complementary CMOS only; every cell is built from NMOS and PMOS transistors or from lower-level cells of the same kind.
- Exactly two opcode-selected operations (ADD and SUB), selected by a 1-bit opcode, as directed by the instructor for this phase.
- Subtraction shall reuse the adder; no separate subtractor.
- Single supply, with every input driven and no floating inputs; one separate testbench cell for each simulated module.
- Pin names are not changed after the cells are connected in higher levels, because renaming breaks connections (guideline Section 13). The resulting supply-pin naming is explained in Section 5.7.

5.7 Pin Names: Proposal versus Implementation

The proposal named the symbol pins A<3:0>, B<3:0>, R<3:0>, VDD and GND. The implemented symbols use the individual bit names A0 to A3, B0 to B3 and R0 to R3 and the lower-case supply

names vdd and gnd. In addition, three library cells use different labels for the supplies. All of them are connected correctly at the slice level, and they are not renamed. Table 4 gives the mapping.

Table 4. Proposal pin names and the pins used in the Cadence implementation.

Proposal / guideline name	Implemented pin	Where
A<3:0>, B<3:0>, R<3:0>	A0–A3, B0–B3, R0–R3	4bit_ALU symbol and testbench
OP, OVF	OP, OVF	unchanged
VDD	vdd	ALU, slice, full adder, NOR2, inverter
VDD	vgs	xor cell (positive supply pin)
GND	gnd	ALU, slice, full adder, NOR2, inverter, xor
GND	vss	nand cell (ground pin)
VDD, VSS on the full adder	vdd, gnd	full_adder (the supply pins of the current cell are vdd and gnd; VSS appeared only in an earlier symbol)

6. System Architecture

6.1 Subtraction by Reuse of the Adder

The central design decision is that subtraction does not require a separate subtractor. In two's complement arithmetic, as treated in standard texts [1], [2], A − B equals A + B' + 1, where B' is the bitwise complement of B. An XOR gate placed in front of each B input therefore passes B unchanged when OP = 0 and inverts it when OP = 1, and feeding OP into the carry-in of the least significant slice supplies the required +1. Four identical one-bit slices are chained so that each carry ripples into the next. The equations were written down before any schematic was drawn; they are given in Table 5 for each bit i from 0 to 3.

Table 5. Design equations (for each bit i = 0 to 3).

Equation	Meaning
B'i = Bi XOR OP	Conditional inversion of B
C0 = OP	Carry-in of slice 0
Pi = Ai XOR B'i	Propagate
Ri = Pi XOR Ci	Result bit
Ci+1 = [(Ai · B'i)' · (Ci · Pi)']'	Carry-out, NAND-NAND form of Ai·B'i + Ci·Pi
Ci+1 = Ai·B'i + Ai·Ci + B'i·Ci	Same carry-out in majority form, which is how the implemented full adder realises it (Section 6.5)
OVF = C4 XOR OP	Carry-out in ADD, borrow in SUB

Written bit by bit with C0 = OP: R0 = A0 ⊕ B'0 ⊕ OP and C1 = A0·B'0 + OP·(A0 ⊕ B'0); R1 = A1 ⊕ B'1 ⊕ C1 and C2 = A1·B'1 + C1·(A1 ⊕ B'1); and likewise for bits 2 and 3, with OVF = C4 ⊕ OP.

6.2 Rationale for OVF = C4 XOR OP

In addition mode, the flag is simply the carry-out of the most significant slice, C4. In subtraction, the adder evaluates A + B' + 1. A carry-out of 1 from this sum means that A was not smaller than B, so the unsigned result is valid and no borrow occurred. A carry-out of 0 means that a borrow took place. Exclusive-ORing C4 with OP inverts the flag in subtraction only, so that OVF equals 1 when the result is invalid in both modes, using a single additional gate.

6.3 Worked Examples

Table 6. Worked examples of the add/subtract rule.

Example	B’ and carry-in	C4	R3..R0	OVF
5 + 3 (OP = 0)	0011, cin 0	0	1000 (8)	0
15 + 1 (OP = 0)	0001, cin 0	1	0000 (0)	1 (carry-out)
5 − 3 (OP = 1)	1100, cin 1	1	0010 (2)	0
3 − 5 (OP = 1)	1010, cin 1	0	1110 (14, that is −2 modulo 16)	1 (borrow)

6.4 Hierarchy

The design is built bottom-up. Each level was built as a schematic, converted into a symbol, and instantiated by the level above, so that every cell is verified once and reused unchanged. Table 7 lists the cells of library jubayer.

Table 7. Cell hierarchy (library jubayer).

Level	Cadence cell	Contents	Pins
Inverter	inverter_1	1 PMOS + 1 NMOS	a, y, vdd, gnd
NAND2	nand	2 PMOS + 2 NMOS	a, b, y, vdd, vss
NOR2	NOR	2 PMOS + 2 NMOS	a, b, y, vdd, gnd
XOR2	xor	3 inverter_1 + 4 PMOS + 4 NMOS	a, b, y, vgs, gnd
Full adder	full_adder	2 xor + 3 nand + 5 inverter_1 + 2 NOR	A, B, Cin, S, Cout, vdd, gnd
Slice	1bit_ALU	1 xor + 1 full_adder	A, B, Cin, OP, R, Cout, vdd, gnd
ALU	4bit_ALU	4 × 1bit_ALU + 1 xor	A0–A3, B0–B3, OP, R0–R3, OVF, vdd, gnd
ALU testbench	4bit_ALU_test	4bit_ALU + 9 vpwl sources + supply + 5 loads	–

6.5 Full Adder

The full adder (cell full_adder) is shown as built in Figure 2. The sum path uses two XOR2 gates: the first forms the propagate signal P = A ⊕ B’ and the second forms the sum S = P ⊕ Cin, which is the result bit R. The carry path realises the majority function. Three NAND2 gates compute (A·B’)’, (A·Cin)’ and (B’·Cin)’, and an inverter after each turns them into the three product terms. A NOR2 gate and an inverter OR the first two products, and a second NOR2 gate and inverter OR that result with the third, giving Cout = A·B’ + A·Cin + B’·Cin. This equals the NAND-NAND form A·B’ + Cin·P of the proposal, because both are the majority function of A, B’ and Cin. In total the full adder has 12 cell instances.

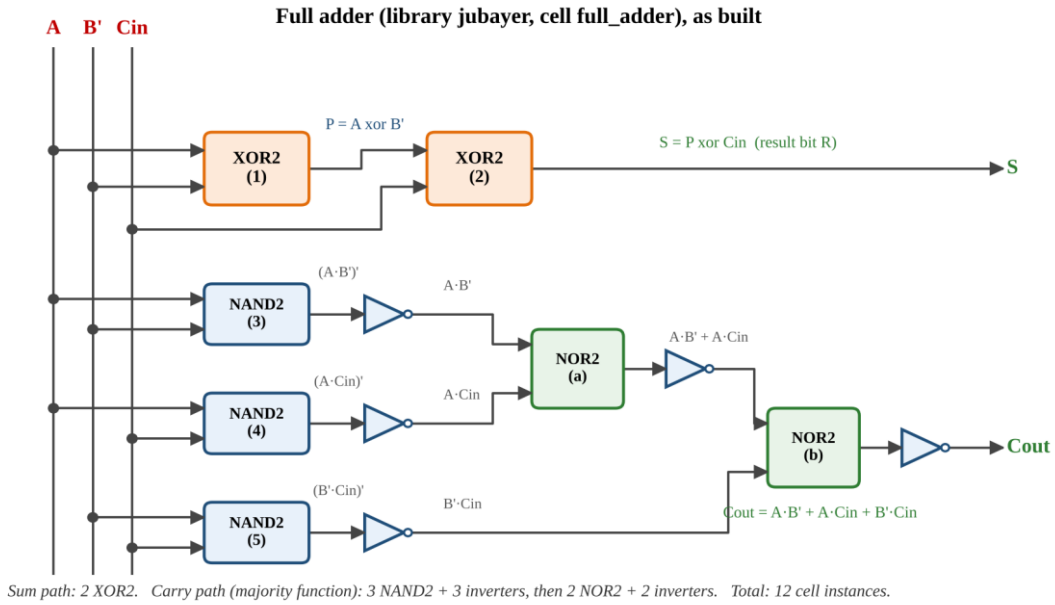


Figure 2. Full adder as built: two XOR2 gates make the sum, three NAND2 gates with inverters and two NOR2 gates with inverters make the majority carry.

Table 8 gives the expected truth table of the full adder, calculated from the equations. It is the reference for the full-adder test cell (Section 8.4).

Table 8. Full adder: expected truth table (calculated).

A	B’	Cin	S	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

6.6 One-Bit Add/Subtract Slice

The slice (library jubayer, cell 1bit_ALU) places one XOR2 on the B input, controlled by OP, and feeds its output B’ into the B input of the full adder (Figure 3). Operand bit A and the carry-in Cin connect directly to the full adder. The sum output S of the full adder is the slice result R, and Cout leaves the slice towards the next stage. The slice has the pins A, B, Cin, OP, R, Cout, vdd and gnd.

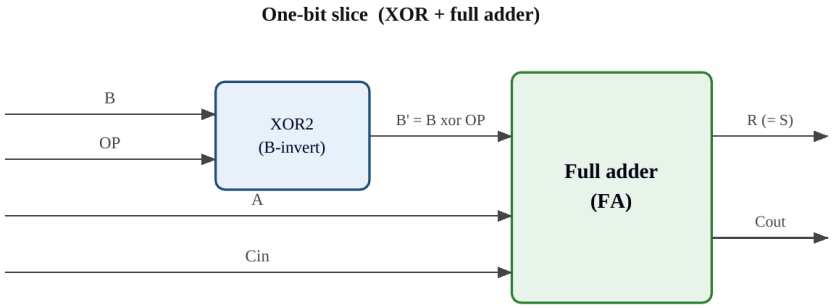


Figure 3. The one-bit slice: an XOR2 conditionally inverts B, then a full adder adds A, B’ and the carry-in.

6.7 Four-Bit ALU

The ALU consists of four slices connected in a ripple-carry chain (Figure 4). OP drives the XOR gate of every slice and is also tied to the carry-in of slice 0. The carry-out of slice 3, C4, is combined with OP in one further XOR2 gate to produce OVF. The symbol of the 4bit_ALU cell exposes exactly the interface defined in Section 5.1.

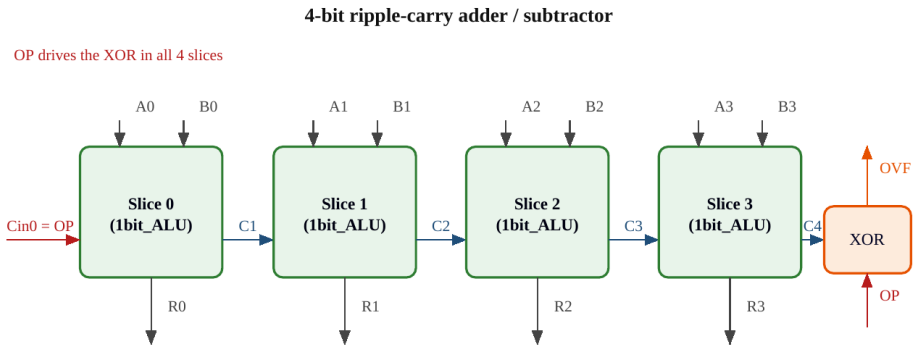


Figure 4. Four slices in a ripple-carry chain, with the carry-in of slice 0 tied to OP and one XOR2 producing OVF.

6.8 Critical Path

The slowest path is the carry rippling from slice 0 to C4. With A = 1111 and B0 toggling in ADD mode, a change at B0 passes through the B-inversion XOR2, then the six gate levels of each full adder carry network (NAND2, inverter, NOR2, inverter, NOR2, inverter) in slices 0, 1 and 2, and finally the sum XOR2 of slice 3. Counting every XOR2 as one gate level, this is 1 + 6 + 6 + 6 + 1 = 20 gate levels to R3. The carry of slice 3 adds six more levels and the OVF XOR2 one more, giving 26 gate levels to OVF.

These are hand counts of gate levels; the measured delays are given in Section 9.4. The earlier draft of this report quoted nine and eleven levels, which belong to the two-level NAND-NAND carry of the proposal; with the full adder as built, the counts above apply. The long path is the reason why the outputs are read at the end of each 10 ns window, after all internal signals have settled.

7. Schematic, Symbol and Cadence Implementation

7.1 Technology and Tools

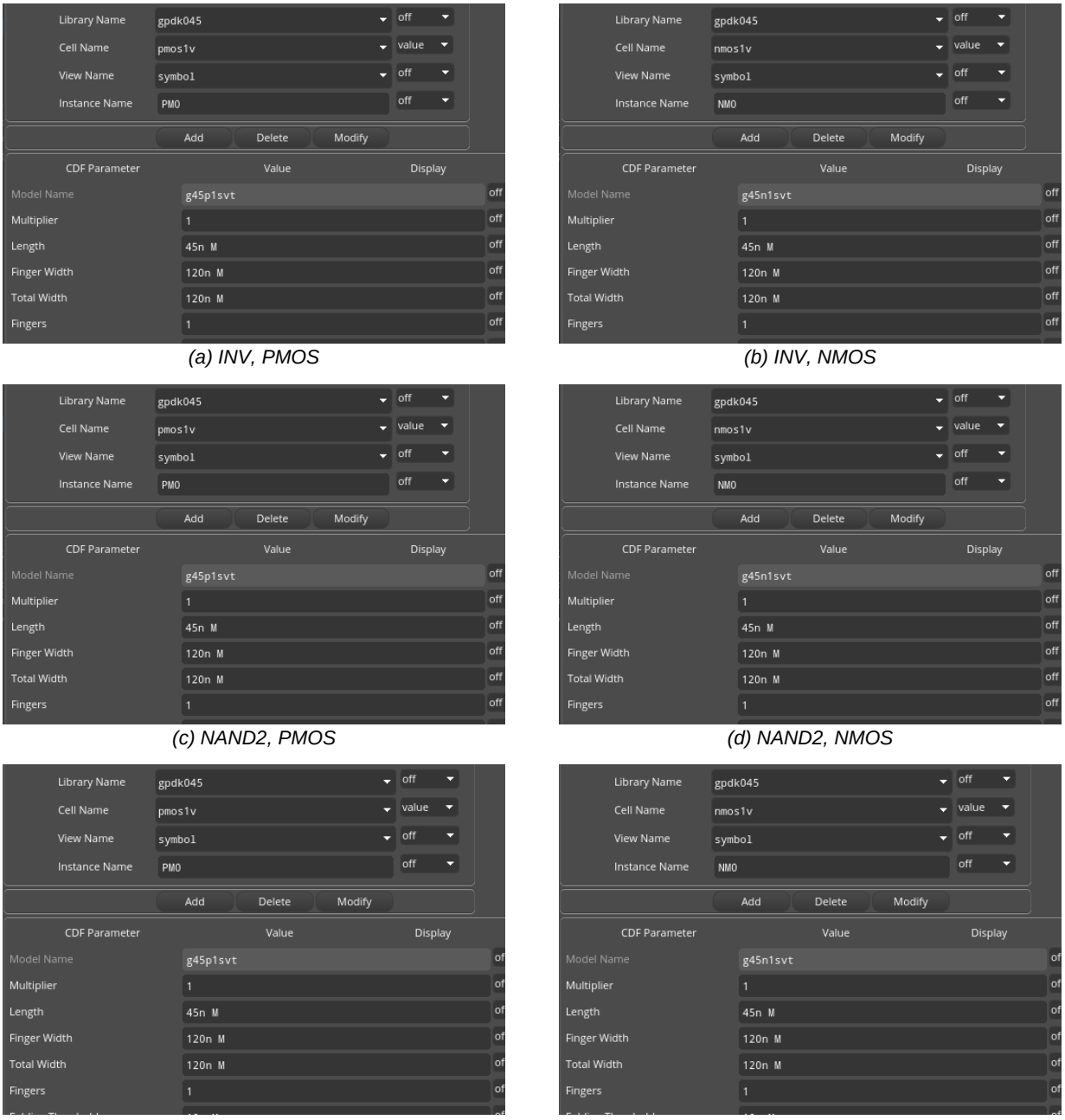
All cells were designed in the gpdk045 (45 nm) process design kit using Cadence Virtuoso for schematic and symbol capture, and were simulated with Cadence Spectre through ADE (Maestro). Waveforms were viewed and measured in the Virtuoso Visualization and Analysis tool. The generated Spectre netlists include the gpdk045 model file, so the kit's own device models are used throughout. Every transistor is an instance of the kit's nmos1v or pmos1v cell. The design library is jubayer, with the cells listed in Table 7.

7.2 Transistor Sizing

Transistor sizes were read from the Property Editor of each leaf cell and confirmed in the generated netlists. Every transistor in the design has the same size: W = 120 nm, L = 45 nm, one finger and multiplier 1 (Table 9, Figure 5). No width ratioing was applied: PMOS and NMOS widths are equal, and series stacks are not widened. This differs from the sizing rule written in the proposal (Appendix B) and has a consequence for speed, discussed in Section 9.6.

Table 9. Transistor sizes of the leaf cells (all transistors: W = 120 nm, L = 45 nm, nf = 1, m = 1).

Cell	Transistors	PMOS model	NMOS model	PMOS / NMOS	Total
inverter_1	PM0, NM0	g45p1svt	g45n1svt	1 / 1	2
nand	PM0, PM1, NM0, NM1	g45p1svt	g45n1svt	2 / 2	4
NOR	PM0, PM1, NM0, NM1	g45p1svt	g45n1svt	2 / 2	4
xor	PM0–PM3, NM0–NM3, and 3 × inverter_1	g45p1svt	g45n1svt	7 / 7	14



(a) INV, PMOS *(b) INV, NMOS*

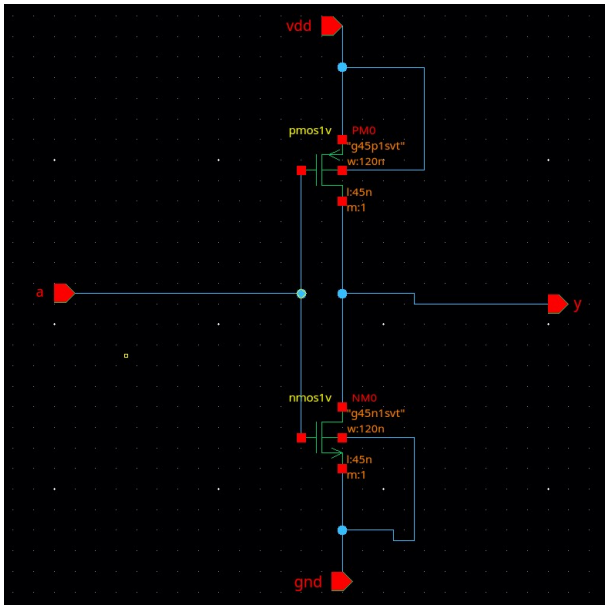
(c) NAND2, PMOS *(d) NAND2, NMOS*

(e) XOR2, PMOS *(f) XOR2, NMOS*

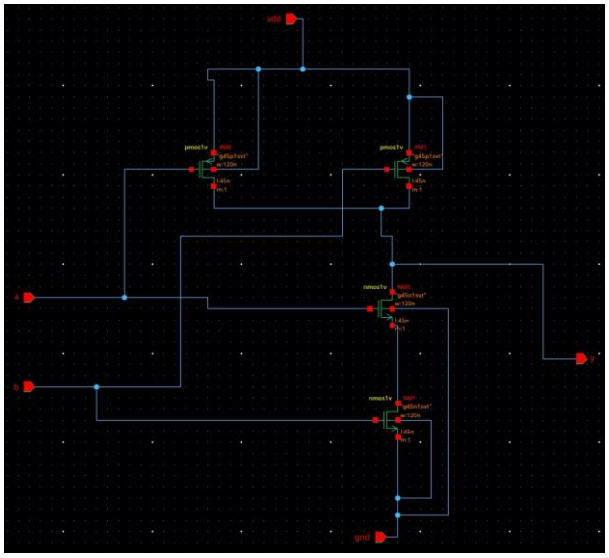
Figure 5. Property Editor values of the transistors of the INV, NAND2 and XOR2 cells: Length 45 nm, Finger Width and Total Width 120 nm, Fingers 1, Multiplier 1.

7.3 Inverter and NAND2

The inverter (cell inverter_1) is one PMOS and one NMOS with common gate input a and output y. The NAND2 (cell nand) has two parallel PMOS transistors between vdd and y and two series NMOS transistors between y and ground (Figure 6). The inverter is used inside the XOR2 and the full adder, and the NAND2 inside the full adder.



(a) inverter_1



(b) nand

Figure 6. Schematics of the inverter and the NAND2 cell.

7.4 NOR2

The NOR2 (cell NOR) has two series PMOS transistors between vdd and y and two parallel NMOS transistors between y and ground. It is used twice in each full adder.

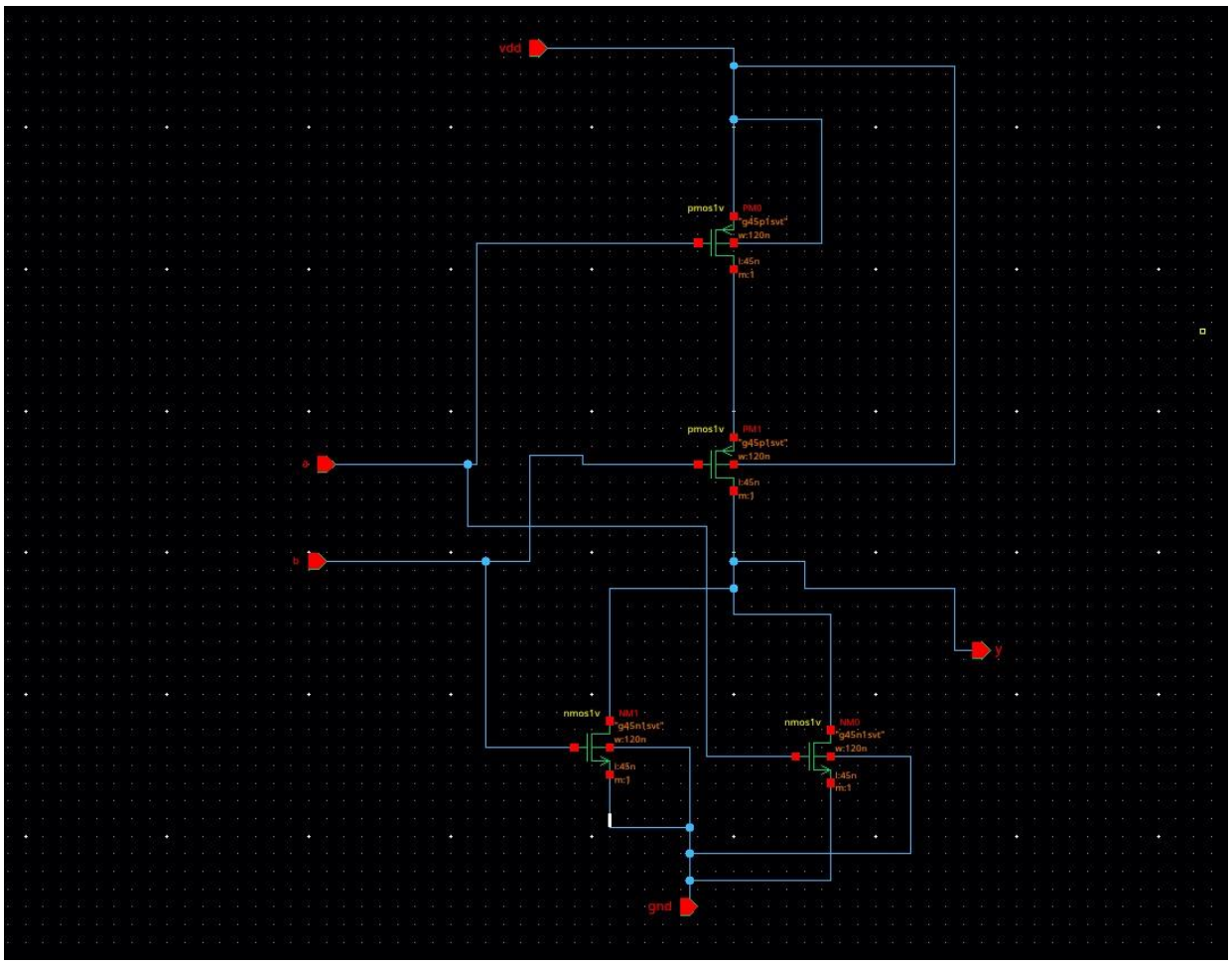


Figure 7. Schematic of the NOR2 cell.

7.5 XOR2

The XOR2 (cell xor) is built as a complementary network fed by the true and complemented inputs. Two input inverters generate a’ and b’, four PMOS and four NMOS transistors form the network, and an output inverter restores y = a ⊕ b. It therefore uses 14 transistors, not the 12 assumed in the proposal. Its schematic and symbol are shown in Figure 8 and Figure 9; the positive supply pin of this cell is labelled vgs.

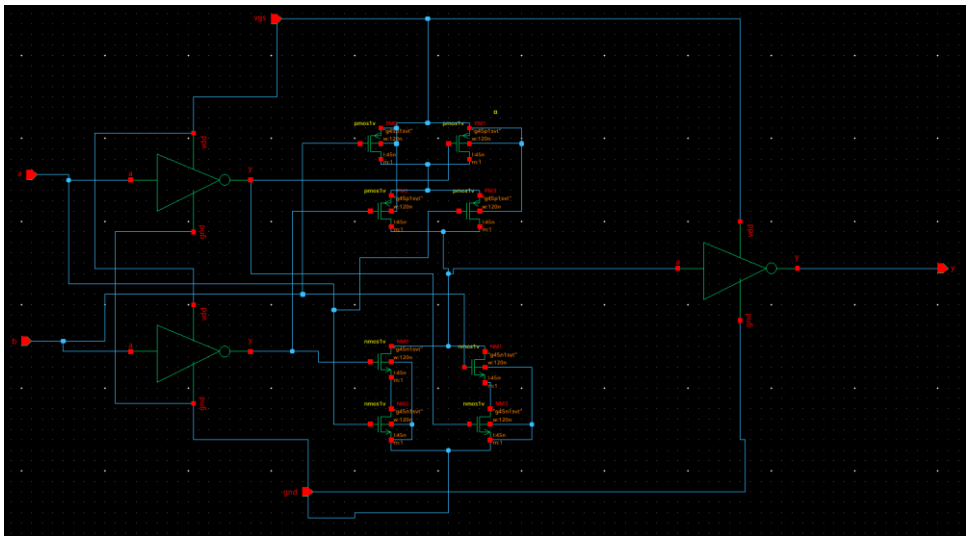


Figure 8. Schematic of the XOR2 cell (library jubayer, cell xor).

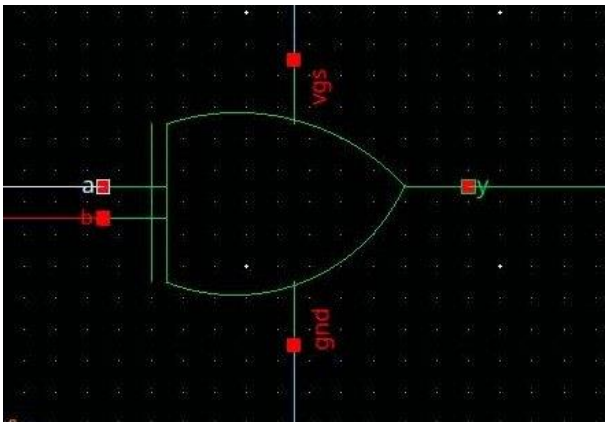


Figure 9. Symbol of the XOR2 cell: inputs a and b, output y, supply pin vgs and ground pin gnd.

7.6 Full Adder

The full adder (cell full_adder) instantiates the cells above as shown in Figure 2 and Figure 10: two xor (I0, I1), three nand (I2, I3, I5), five inverter_1 (I6–I10) and two NOR (I11, I12). Its pins are A, B, Cin, S, Cout, vdd and gnd. The xor instances receive vdd on their vgs pin and the nand instances receive gnd on their vss pin.

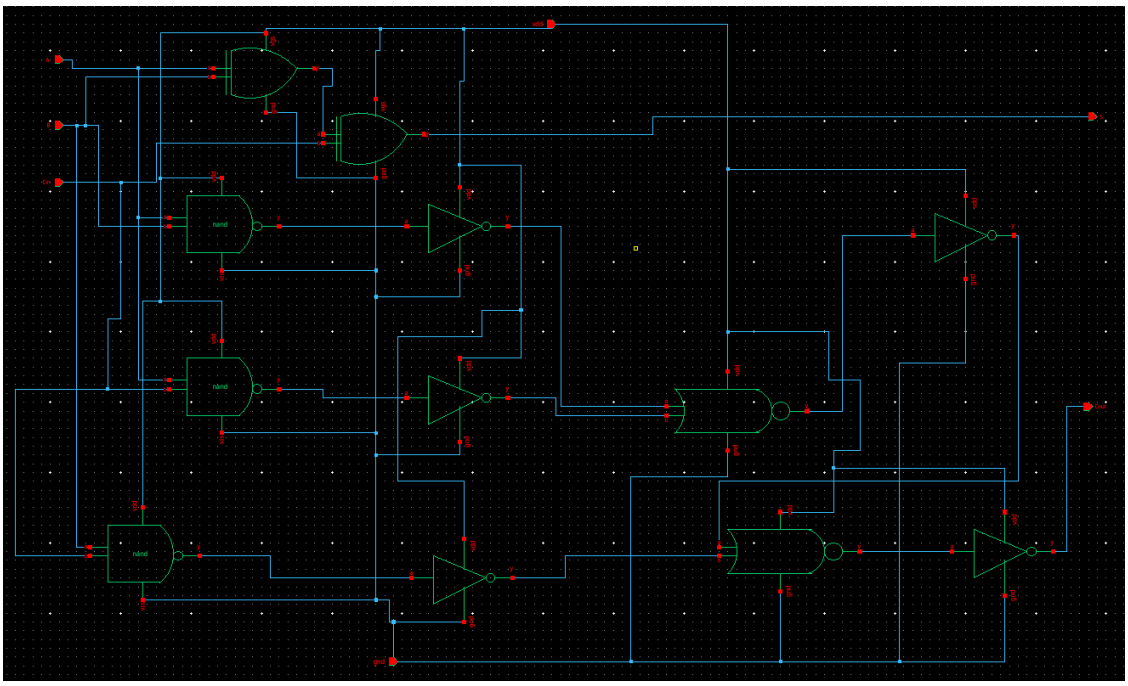


Figure 10. Schematic of the full adder (library jubayer, cell full_adder).

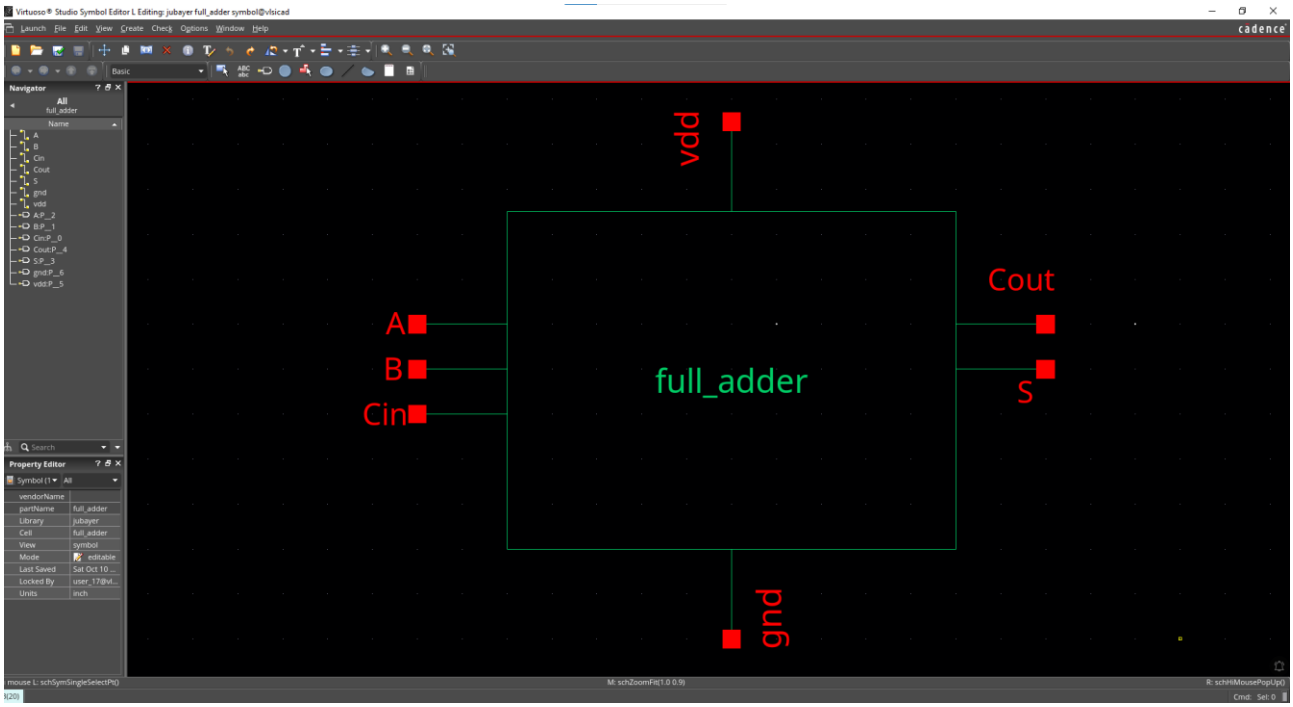


Figure 11. Symbol of the full adder.

7.7 One-Bit Slice

In the slice schematic (Figure 12), the input B goes into the XOR2 together with OP. The XOR2 output y is the inverted-or-not B' and feeds the full adder's B input. A and Cin go straight to the full adder. The full adder's S output is the slice result R, and Cout leaves the slice. A symbol with A, B, Cin and OP on the left, Cout and R on the right, vdd on top and gnd below was created in the symbol editor (Figure 13).

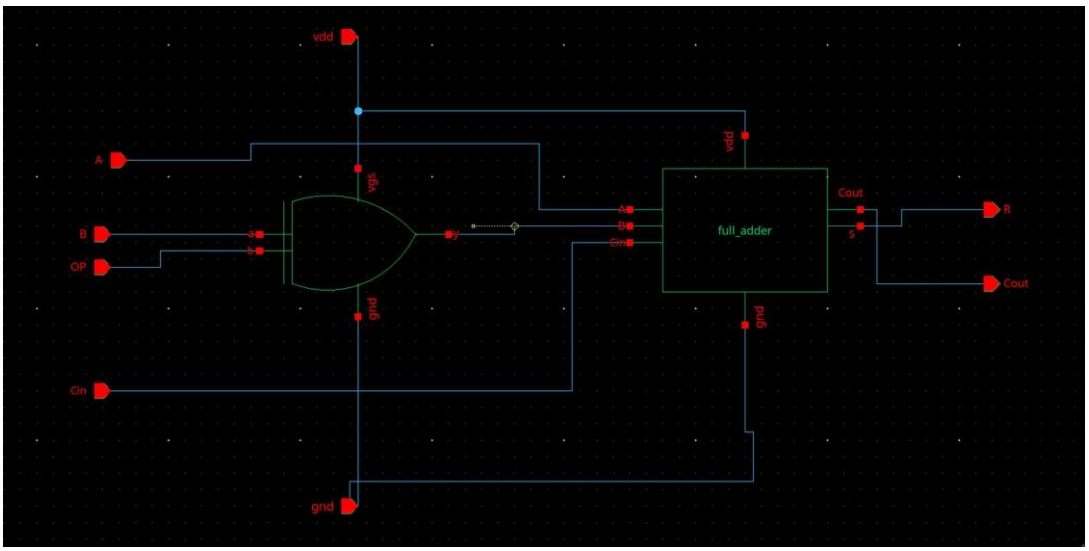


Figure 12. Slice schematic (library jubayer, cell 1bit_ALU). Pins: A, B, Cin, OP, R, Cout, vdd, gnd.

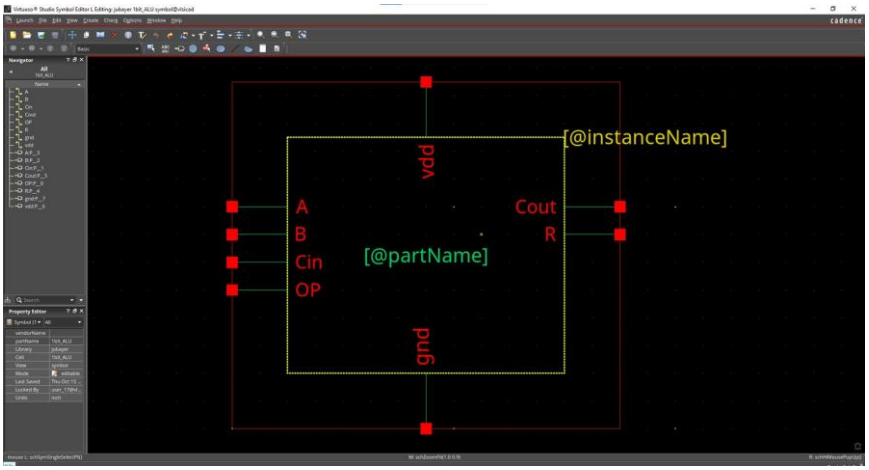
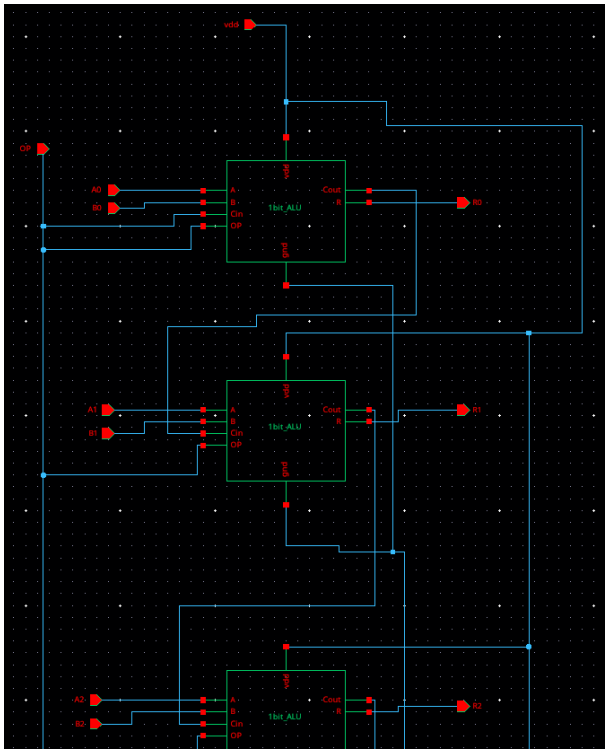


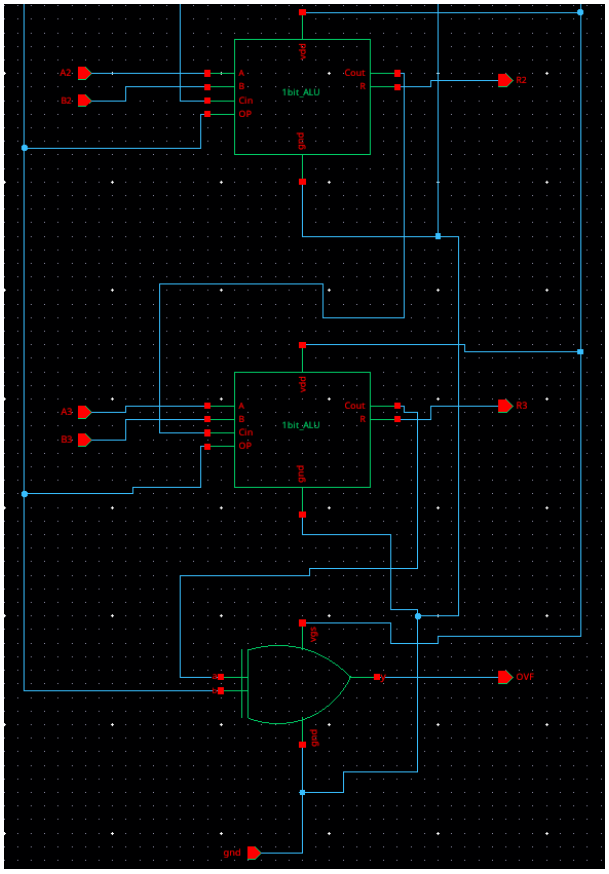
Figure 13. Symbol editor: A, B, Cin, OP on the left; Cout, R on the right; vdd on top, gnd below.

7.8 Four-Bit ALU

The 4bit_ALU schematic instantiates four copies of the slice symbol and the OVF XOR2 (Figure 14). Its symbol (Figure 15) presents A0 to A3, B0 to B3 and OP as inputs, OVF and R0 to R3 as outputs, and vdd and gnd as supplies, which is exactly the interface defined in Section 5.1.



(a) upper half



(b) lower half

Figure 14. Schematic of the 4bit_ALU cell, shown in two halves.

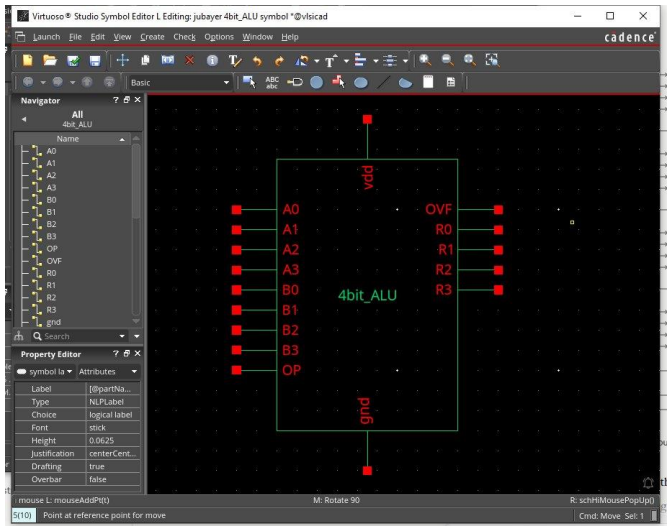


Figure 15. The 4bit_ALU symbol: A0–A3, B0–B3 and OP in; OVF and R0–R3 out; vdd and gnd.

7.9 Supply Pins, Bulk Connections and Checks

The cells use different labels for the supply pins, as listed in Table 10. They are all connected correctly: at the slice level the vdd net feeds the vgs pin of the xor and the vdd pins of the full adder, and the gnd net feeds the gnd pins of both. Inside the full adder, the nand cells receive gnd on vss. The pins were not renamed, because renaming breaks the connections in the higher levels.

Table 10. Supply-pin labels of the library cells.

Cell	Signal pins	Positive supply pin	Ground pin
inverter_1	a, y	vdd	gnd
nand	a, b, y	vdd	vss
NOR	a, b, y	vdd	gnd
xor	a, b, y	vgs	gnd
full_adder	A, B, Cin, S, Cout	vdd	gnd
1bit_ALU	A, B, Cin, OP, R, Cout	vdd	gnd
4bit_ALU	A0–A3, B0–B3, OP, R0–R3, OVF	vdd	gnd

In every leaf cell the bulk terminal of each NMOS transistor is tied to the ground pin and the bulk terminal of each PMOS transistor to the positive supply pin, as the netlists show.

8. Simulation and Verification

8.1 Strategy

Verification was performed at two levels. The one-bit slice was tested exhaustively, which is feasible because it has only four inputs, and the full ALU was tested with a directed set of 13 cases chosen to exercise ordinary operation and every boundary condition of the specification. For every case, the expected result was computed from the specification before simulation, and the simulated value was read from the output waveform near the end of its 10 ns window, after the circuit had settled. In all ALU and slice simulations each output was loaded with a 10 fF capacitor to ground. The conditions are listed in Table 3.

8.2 Slice Testbench

The 1bit_ALU symbol was placed in a test schematic (Figure 16) and driven through all 16 combinations of A, B, Cin and OP, 10 ns per combination, in a single 160 ns transient run. The stimuli come from four vpulse sources (Table 11), the supply is the vdc source V4 at 1.8 V, and a 10 fF capacitor loads each of the outputs Cout and R. The expected values follow R = A XOR (B XOR OP) XOR Cin and Cout = majority(A, B XOR OP, Cin).

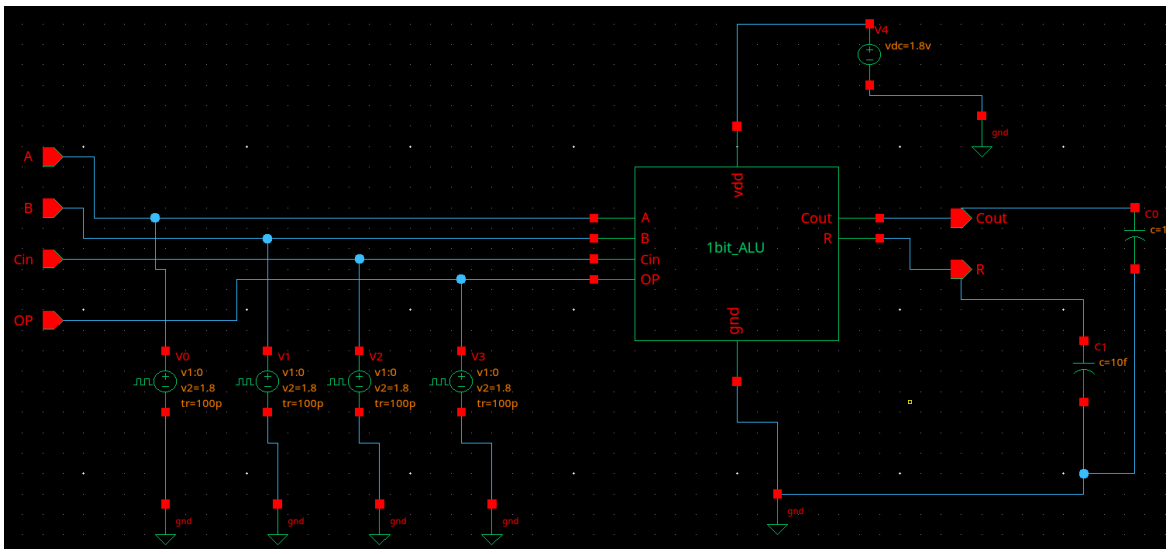


Figure 16. Slice testbench: four vpulse sources (V0–V3, 0 to 1.8 V, tr = 100 ps), supply V4 = 1.8 V and 10 fF loads on Cout and R.

Table 11. Slice testbench sources.

Source	Drives	Behaviour
V0	A	Toggles every 10 ns
V1	B	Toggles every 20 ns
V2	Cin	Toggles every 40 ns
V3	OP	Low until 80 ns, high from 80 ns
V4	supply	vdc = 1.8 V between vdd and gnd

Results: exhaustive test of the slice

All 16 input combinations produced the expected result; expected and simulated values were identical in every row of Table 12. Figure 17 shows that R goes high whenever an odd number of the signals A, B’ and Cin is high, and that Cout goes high whenever at least two of them are high, as the equations require.

Table 12. One-bit slice: expected versus simulated results (16 of 16 pass).

Case	Window (ns)	OP	Cin	B	A	Expected R, Cout	Simulated R, Cout	Result
1	0–10	0	0	0	0	0, 0	0, 0	PASS
2	10–20	0	0	0	1	1, 0	1, 0	PASS
3	20–30	0	0	1	0	1, 0	1, 0	PASS
4	30–40	0	0	1	1	0, 1	0, 1	PASS
5	40–50	0	1	0	0	1, 0	1, 0	PASS
6	50–60	0	1	0	1	0, 1	0, 1	PASS
7	60–70	0	1	1	0	0, 1	0, 1	PASS
8	70–80	0	1	1	1	1, 1	1, 1	PASS
9	80–90	1	0	0	0	1, 0	1, 0	PASS
10	90–100	1	0	0	1	0, 1	0, 1	PASS
11	100–110	1	0	1	0	0, 0	0, 0	PASS
12	110–120	1	0	1	1	1, 0	1, 0	PASS
13	120–130	1	1	0	0	0, 1	0, 1	PASS
14	130–140	1	1	0	1	1, 1	1, 1	PASS
15	140–150	1	1	1	0	1, 0	1, 0	PASS
16	150–160	1	1	1	1	0, 1	0, 1	PASS

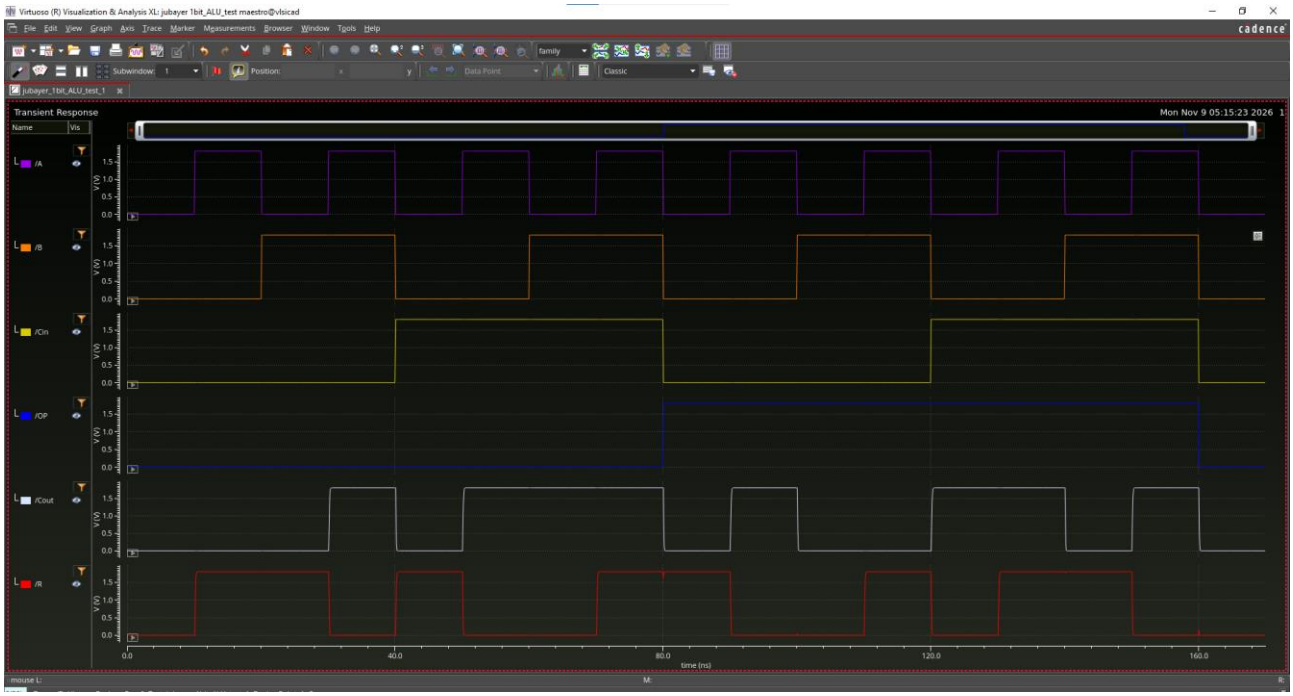


Figure 17. All six signals of the slice testbench (A, B, Cin, OP, R, Cout) over 160 ns.

8.3 Hand Calculation of Expected Values

The expected values in Table 12 and Table 13 were calculated from the equations of Section 6.1 before simulation. As an independent check that holds in every case and for both operations, R0 equals A0 XOR B0. This follows from the equations, because R0 = A0 XOR (B0 XOR OP) XOR OP, and the two occurrences of OP cancel.

8.4 Full Adder and XOR2 Test Cells

A separate test cell was also created for the full adder (cell full_adder_test): three vpulse sources drive A, B and Cin with a 1 ns delay and 1 ns edges (A toggles every 10 ns, B every 20 ns, Cin every 40 ns, so that all eight combinations of Table 8 occur), and a 1.8 V vdc source supplies the cell. The XOR2 was tested in a similar test cell with two vpulse sources and a 1.8 V vdc source (Figure 18). The waveforms and pass/fail tables of these two cell-level tests are not included in this report; the slice test of Section 8.2 exercises the full adder and the XOR2 together over every input pattern one bit position can see.

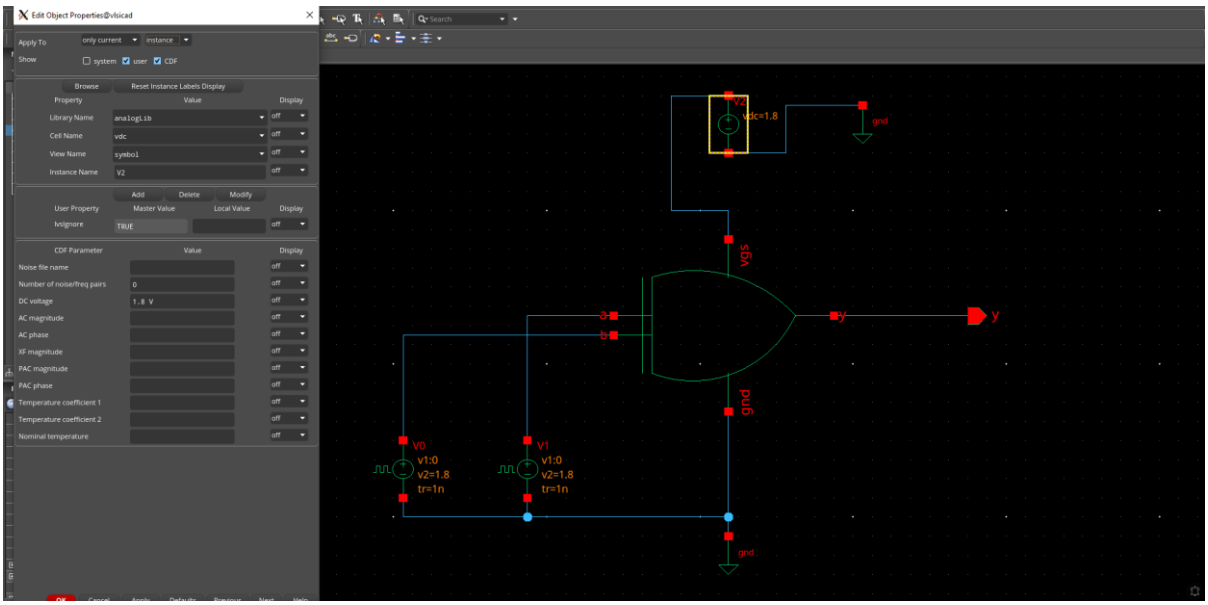


Figure 18. XOR2 test cell: vpulse sources V0 and V1 drive a and b, and the vdc source V2 = 1.8 V supplies the cell (Edit Object Properties window shown).

8.5 ALU Testbench

The ALU testbench (cell 4bit_ALU_test, Figure 19 and Figure 20) applies the nine inputs A0 to A3, B0 to B3 and OP from nine piecewise-linear (vpwl) voltage sources, V0 to V8, with the supply provided by a tenth source, V9, at 1.8 V. Five 10 fF capacitors load the outputs R0 to R3 and OVF. Each level change in a source is entered as two points, the old level at the change time and the new level 0.1 ns later, so that all edges are clean. Case k occupies the interval from (k − 1) × 10 ns to k × 10 ns, and the transient analysis was run to 150 ns, which covers all 13 cases (0 to 130 ns). The complete stimulus definition, together with the property-editor screenshots of every source, is given in Appendix A.

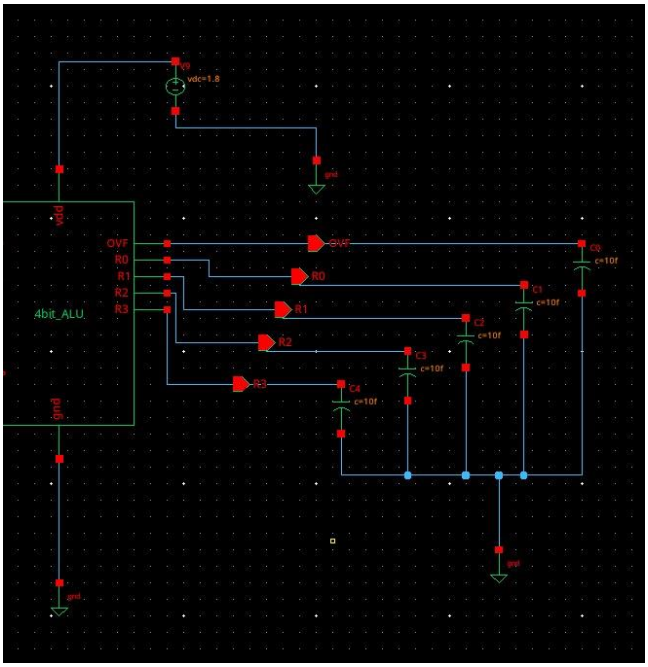


Figure 19. Output side of the testbench: 1.8 V supply (V9) and a 10 fF load on R0–R3 and OVF.

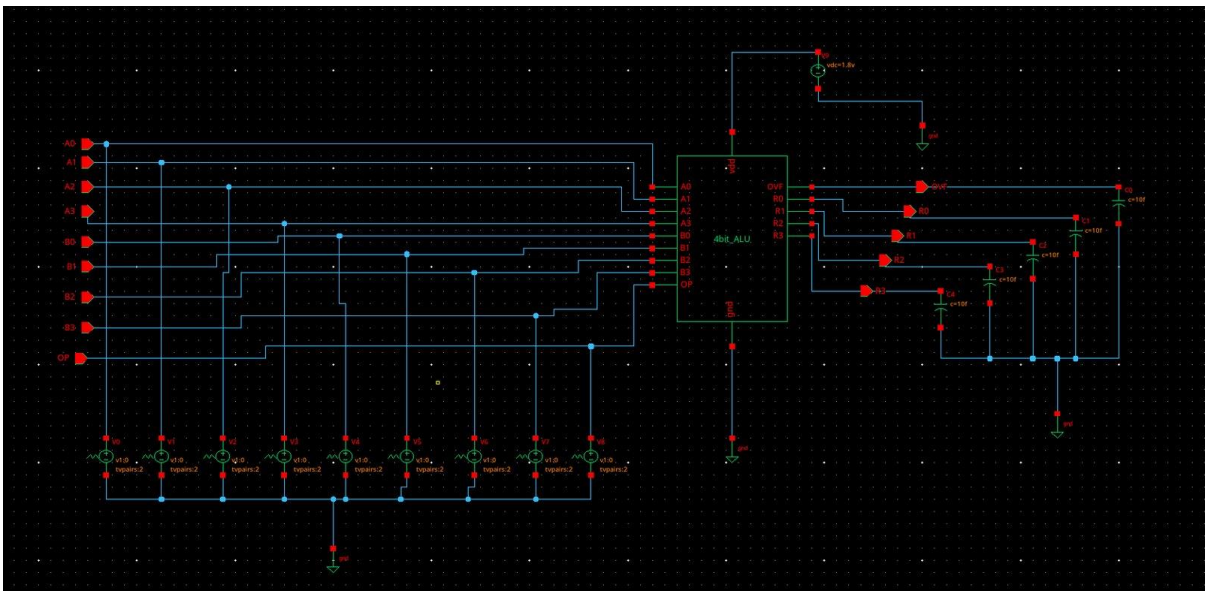


Figure 20. 4bit_ALU_test: nine vpwl sources (V0–V8) drive A0–A3, B0–B3 and OP; V9 is the supply; 10 fF capacitors load the outputs.

8.6 ALU Results

Table 13 compares the expected and simulated results for all 13 cases. Figure 21 shows the simulator output as plotted, and Figure 22 shows the annotated version with the input stimulus, the opcode and all outputs on one time axis, with the 10 ns window of every case marked and numbered. OVF is high in cases 5 and 6 (carry-out in ADD) and in cases 12 and 13 (borrow in SUB, from 110 ns onwards).

Table 13. Four-bit ALU: expected (Exp.) versus simulated (Sim.) results, 13 of 13 pass. Cases 5, 6, 12 and 13 are the boundary cases.

No.	Window (ns)	OP	A	B	Operation	Exp. R3..R0	Exp. OVF	Sim. R3..R0	Sim. OVF	Result
1	0–10	0	0101	0011	5 + 3 = 8	1000	0	1000	0	PASS
2	10–20	0	0000	0000	0 + 0 = 0	0000	0	0000	0	PASS
3	20–30	0	0111	0001	7 + 1 = 8	1000	0	1000	0	PASS
4	30–40	0	1010	0101	10 + 5 = 15	1111	0	1111	0	PASS
5	40–50	0	1111	0001	15 + 1 = 0 (carry)	0000	1	0000	1	PASS
6	50–60	0	1111	1111	15 + 15 = 14 (carry)	1110	1	1110	1	PASS
7	60–70	1	0111	0010	7 − 2 = 5	0101	0	0101	0	PASS
8	70–80	1	0101	0011	5 − 3 = 2	0010	0	0010	0	PASS
9	80–90	1	0111	0111	7 − 7 = 0	0000	0	0000	0	PASS
10	90–100	1	1111	0000	15 − 0 = 15	1111	0	1111	0	PASS
11	100–110	1	1000	0001	8 − 1 = 7	0111	0	0111	0	PASS
12	110–120	1	0011	0101	3 − 5 = 14 (borrow)	1110	1	1110	1	PASS
13	120–130	1	0000	0001	0 − 1 = 15 (borrow)	1111	1	1111	1	PASS

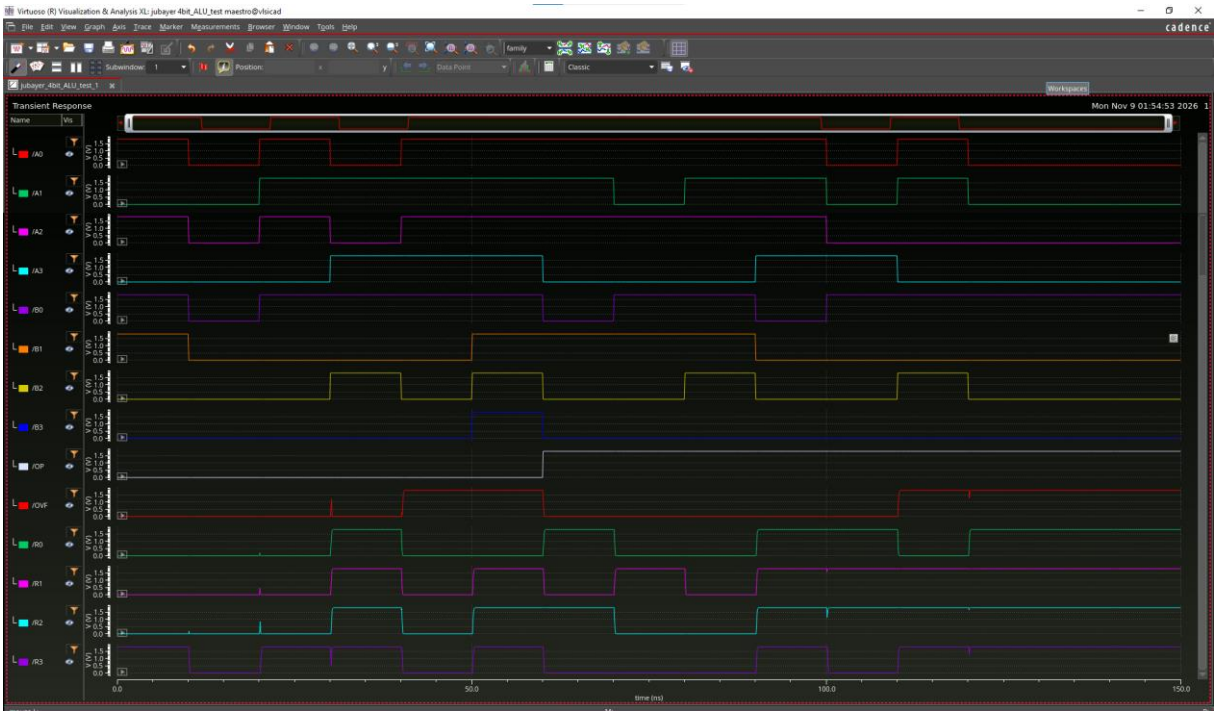


Figure 21. Simulator output of the ALU testbench over 0–150 ns in the waveform viewer: A0–A3, B0–B3, OP, OVF and R0–R3 on a common time axis (stacked from two scrolled screenshots of the same run).

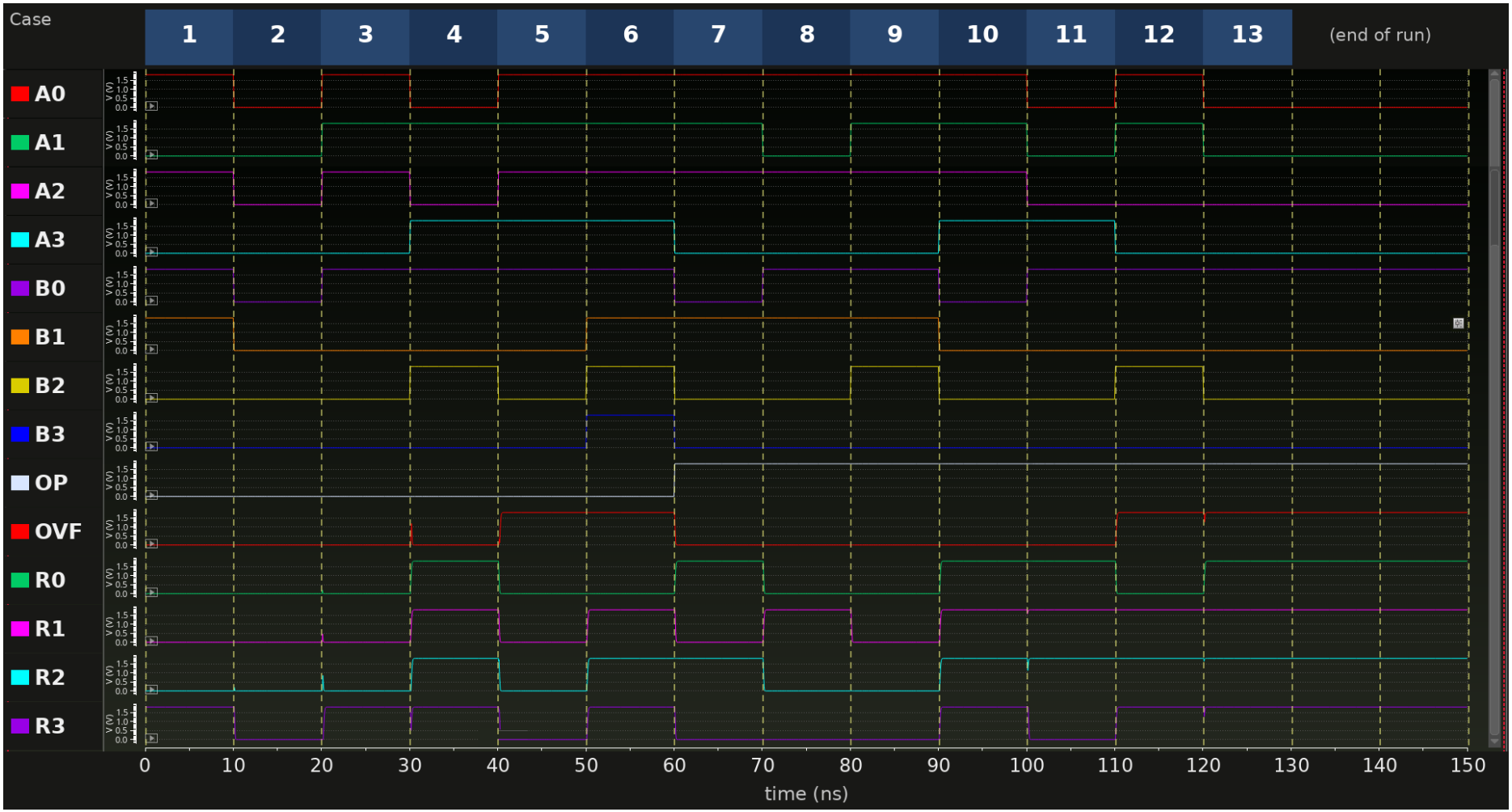


Figure 22. ALU transient simulation over 0–150 ns: A0–A3, B0–B3, OP, OVF and R0–R3 on separate strips with a common time axis. Case numbers 1–13 mark the 10 ns windows (dashed lines every 10 ns); each result is read at the end of its window. The case numbers and signal-name labels were added to the simulator output, and the figure is stacked from two scrolled screenshots of the same run.

8.7 Analysis

All 13 cases agree with the expected values for both the result and the flag. The cases were selected to cover the specification rather than to sample it at random. Zero operands are tested in case 2, and all-ones operands in cases 5, 6 and 10. Wrap-around in addition is tested by 15 + 1 = 0 and 15 + 15 = 14, both with OVF = 1, which confirms that the flag reports the carry-out of the most significant slice. Subtraction with equal operands is tested in case 9, where the result is zero and no borrow is raised, which demonstrates that the +1 supplied through the carry-in works correctly, since a missing carry-in would give 1111 with a spurious borrow. Borrow itself is tested in cases 12 and 13, where the results 1110 and 1111 are the two's complement representations of −2 and −1 and OVF is correctly set.

A useful independent check holds in every case and for both operations: R0 equals A0 XOR B0. This follows from the equations, because R0 = A0 XOR (B0 XOR OP) XOR OP, and the two occurrences of OP cancel. Every simulated R0 satisfies this relation.

Short glitches appear on the outputs at the start of some windows when several inputs change at the same instant. These are static hazards, which are normal in static CMOS combinational logic with reconvergent paths. They die out well before the end of each 10 ns window, which is where the results are read, so they do not affect the verdicts above. The annotated waveform of Figure 22 was also checked against Table 13 by reading the level of all 14 signals at the end of each of the 13 windows from the picture; every level agrees with the table.

8.8 Issues Encountered and Resolutions

Several errors were found during the build, understood, and corrected before the final results were produced. They are summarised in Table 14, and the screenshots that document them are given in Appendix C.

Table 14. Problems found and fixed during the build.

Issue	Cause and consequence	Resolution
Slice wiring	The first slice put the XOR on OP and A and sent B straight to the adder, so the wrong operand would be inverted.	XOR moved to B and OP, with its output feeding the adder's B input.
Load capacitors	A capacitor was placed in series with an output instead of from the output to ground, and one wire was attached to the pin outline instead of its connection point, so the capacitor was not electrically connected.	Each capacitor connected from the output net to ground, with a junction dot at the join.
vpwl source V5 (B1)	Voltage 5 was left at its default of 0 V instead of 1.8 V, so B1 would have dropped out of its high interval.	Entry corrected to 1.8 V.
vpwl source V2 (A2)	Time 11 was entered as 100 ns, equal to Time 10. Spectre stopped with error CMI-2204 because PWL times must strictly increase.	Time 11 changed to 100.1 ns; the full run then completed.

The Spectre log excerpt for the V2 error is reproduced below. The error disappeared after the entry was corrected, and the full run completed.

```
ERROR (CMI-2204): `input.scs':156, V2: In the PWL waveform, time 100 ns, calculated using expression `1e-07' at index 20,
is less than the previously specified time 100 ns. ...
spectre terminated prematurely due to fatal error.
```

9. Performance Analysis

This section evaluates the design by area, output voltage levels, settling, propagation delay and power. Where a quantity was not measured, no value is given.

9.1 Area (Transistor Count)

No layout was made, so the transistor count is used as the area estimate. The counts of Table 15 were taken by counting the transistor instances in the netlist of each cell. The cells down to the full adder were counted directly (58 transistors for the full adder: 2 × 14 + 3 × 4 + 5 × 2 + 2 × 4); the slice and ALU counts are derived by multiplying out the hierarchy. All transistors have W = 120 nm, so the total channel width of the 302-transistor ALU is 302 × 120 nm = 36.24 µm.

Table 15. Cell instances and transistor counts per hierarchy level (as built).

Level	Contents	Cell instances	Transistors
Inverter	1 PMOS + 1 NMOS	1	2
NAND2 / NOR2	2 PMOS + 2 NMOS	1	4
XOR2	3 inverters + 4 PMOS + 4 NMOS	4	14
Full adder	2 XOR2 + 3 NAND2 + 5 inverters + 2 NOR2	12	58
One-bit slice	XOR2 + full adder	13	72 (derived)
4-bit ALU	4 slices + OVF XOR2	53	302 (derived)

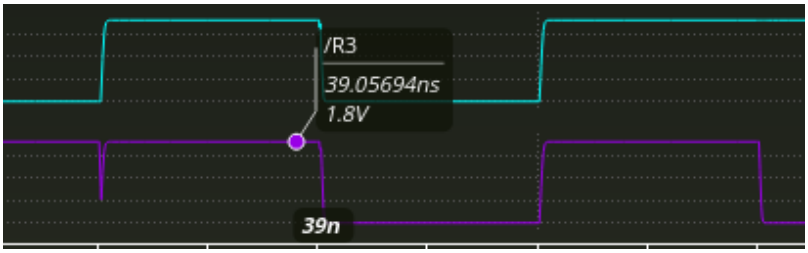
The counts differ from the 12, 36, 48 and 204 transistors of the proposal, because the real XOR2 uses an inverter on each input and one on its output, and the real full adder realises its carry with inverters and NOR gates (Appendix B).

9.2 Output Voltage Levels

The settled output levels were read with markers in the waveform viewer (Figure 23, Table 16). At logic 1 the outputs reach the supply voltage (the marker display shows 1.8 V), and at logic 0 the outputs sit at 141 nV (OVF) and 374 nV (R3), essentially ground. The outputs therefore swing fully between the rails, as expected for static CMOS with no ratioed logic.

Table 16. Output levels read from the transient waveform with a 10 fF load and a 1.8 V supply.

Signal	Time (ns)	Case and expected logic level	Level read	Quantity
R3	39.057	Case 4 (10 + 5 = 15): R3 = 1	1.8 V	VOH
OVF	49.115	Case 5 (15 + 1): OVF = 1	1.8 V	VOH
OVF	9.075	Case 1 (5 + 3): OVF = 0	141.32 nV	VOL
R3	19.02	Case 2 (0 + 0): R3 = 0	374.25 nV	VOL



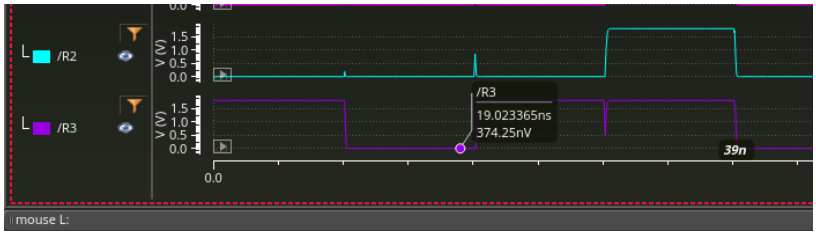
(a) R3 at 39.057 ns: 1.8 V



(b) OVF at 49.115 ns: 1.8 V



(c) OVF at 9.075 ns: 141.32 nV



(d) R3 at 19.02 ns: 374.25 nV

Figure 23. Marker readings of the settled output levels (cropped from the waveform viewer).

The output current was not measured. The only load is the 10 fF capacitor, so in the steady state no current flows into the load, and the supply current was not extracted.

9.3 Settling

In every case the outputs reached their final level before the end of the 10 ns window in which they are read, and the short glitches at the start of some windows (Section 8.7) were gone by then. This shows that the settling time of the ALU, including the full carry ripple, is shorter than 10 ns at this supply and load. The worst-case propagation delay of 275.7 ps (Section 9.4) is far below the 10 ns window.

9.4 Propagation Delay

The propagation delay was measured on the worst-case path, the carry ripple through all four slices, in a copy of the ALU testbench (cell 4bit_ALU_test_delay, Figure 24). The operands were A = 1111, B3 to B1 = 0 and OP = 0, so that a change at B0 ripples through every slice, and B0 was pulsed from 0 to 1.8 V (delay 5 ns, 100 ps edges, width 20 ns, period 40 ns). With A = 1111, raising B0 changes the sum from 1111 to 0000, so R3 falls and OVF rises, and lowering B0 does the reverse (Figure 25). The conditions are those of Table 3: a 1.8 V supply, a 10 fF load on every output and 27 °C.

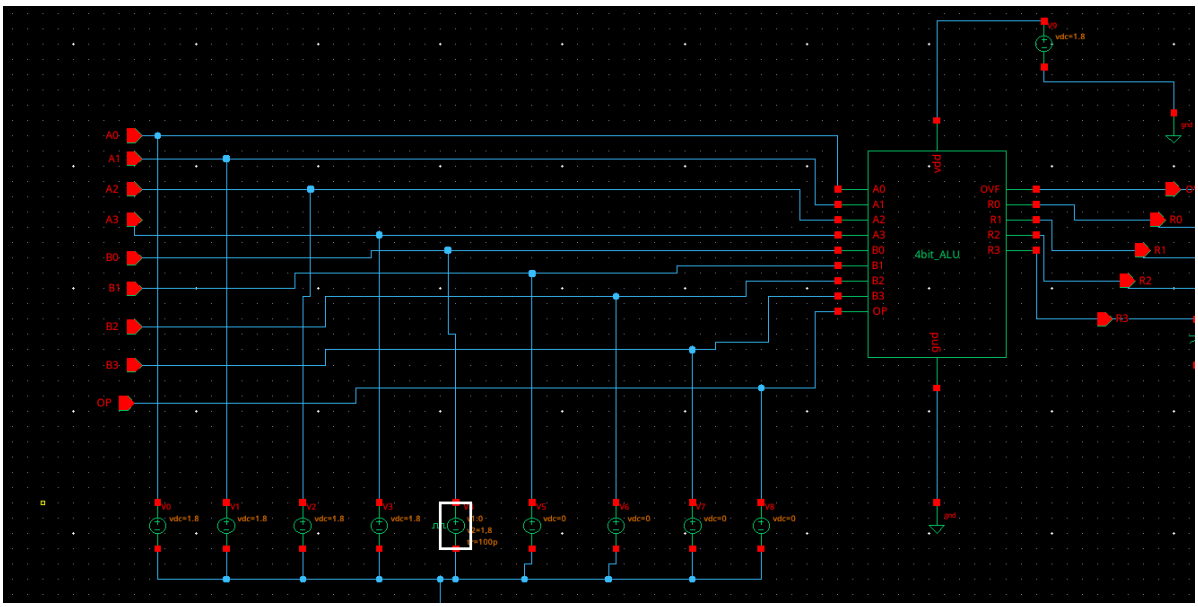


Figure 24. Delay testbench (copy of 4bit_ALU_test): A0–A3 are held at 1.8 V and B1–B3 and OP at 0 V with vdc sources, B0 is driven by a vpulse source, and V9 supplies 1.8 V.



Figure 25. B0, R3 and OVF in the delay testbench over 50 ns. After B0 rises at about 5 ns, R3 falls and OVF rises; after B0 falls at about 25 ns, R3 rises and OVF falls.

Each delay is the time between the 50 percent point (0.9 V) of B0 and the 50 percent point of the output, found with the calculator function cross() of the waveform viewer (Figure 26). The four results are given in Table 17.

Table 17. Propagation delays of the carry path (A = 1111, B = 0000 to 0001, OP = 0, 1.8 V supply, 10 fF load).

No.	B0 edge	Path and output edge	Delay (ps)	Remark
1	rising (about 5 ns)	B0 to R3, R3 falling	216.9	
2	rising (about 5 ns)	B0 to OVF, OVF rising	269.7	
3	falling (about 25 ns)	B0 to R3, R3 rising	261.0	
4	falling (about 25 ns)	B0 to OVF, OVF falling	275.7	Worst case

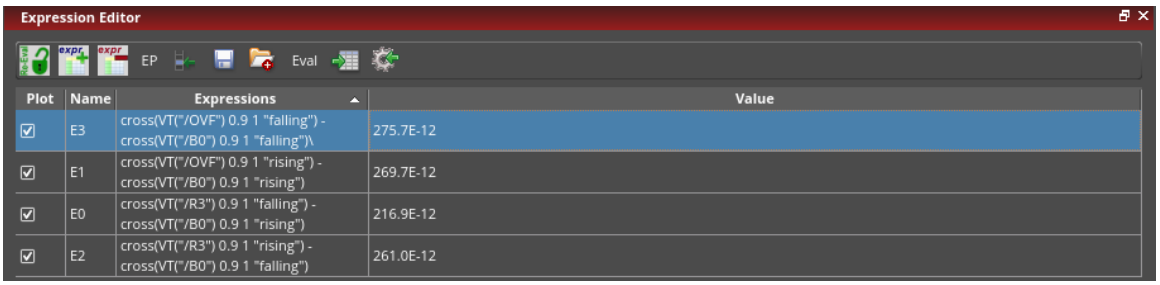


Figure 26. Calculator results for the four delays (E0: B0 rising to R3 falling; E1: B0 rising to OVF rising; E2: B0 falling to R3 rising; E3: B0 falling to OVF falling), in seconds.

The worst-case delay is 275.7 ps, from the falling edge of B0 to the falling edge of OVF. The delay to OVF is longer than the delay to R3 for both edges of B0, as expected from the structural count of Section 6.8 (about 26 gate levels to OVF against about 20 to R3). The measurement with OP switching, which needs A = 0111 and B = 0001 because A = B = 0000 gives no output transition, was not run.

9.5 Power Consumption

The average power was computed as VDD multiplied by the magnitude of the average supply current of source V9 over the 13-case run of the ALU testbench, from 0 to 130 ns. The calculator expression average(clip(IT("/V9/PLUS") 0 130n)) gives −5.696 µA (Figure 27); the negative sign only means that the source delivers current. The average power is therefore 1.8 V × 5.696 µA ≈ 10.25 µW. This value includes the switching and leakage current for this particular stimulus at a 1.8 V supply and a 10 fF load, and it is not a general power rating.

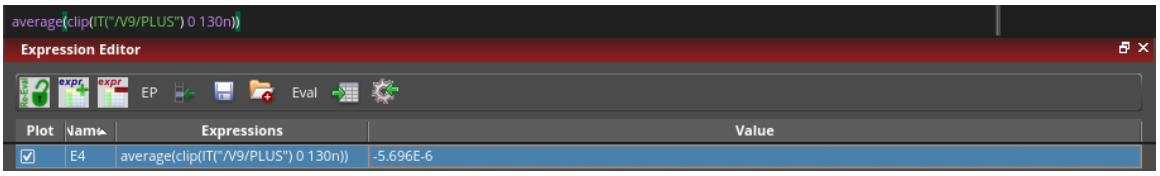


Figure 27. Average supply current of V9 over 0–130 ns (calculator expression E4): −5.696 µA.

9.6 Supply Voltage and Limitations

The supply voltage is 1.8 V, about twice the nominal value of the 45 nm kit (Section 5.4). Operation at the nominal supply was not simulated, so the delay and power values of Sections 9.4 and 9.5 were obtained at this elevated supply and do not represent nominal operation. In addition, every transistor has the minimum length and the same width of 120 nm, with no extra width on the PMOS devices or on series stacks. With equal widths the PMOS transistors are weaker than the NMOS ones, so rising output transitions and the series stacks of the NAND2 and XOR2 networks are slower than they would be in a ratioed design. This does not change the logic function that was verified, but it affects the measured delay. The model file section used in the netlists is mc, with the Maestro corner nom.

9.7 Summary of Performance Results

Table 18. Performance parameters and their status.

Parameter	Result	Status
Function, one-bit slice	16 of 16 cases pass	Measured
Function, 4-bit ALU	13 of 13 cases pass	Measured
Area	302 transistors, 36.24 µm total channel width	Counted (derived)
Output high level (VOH)	1.8 V (R3 and OVF)	Measured
Output low level (VOL)	141.32 nV (OVF); 374.25 nV (R3)	Measured
Settling	Within the 10 ns window; worst-case delay 275.7 ps	Measured
Propagation delay (worst case)	275.7 ps (B0 falling to OVF falling)	Measured
Average power	≈ 10.25 µW (5.696 µA average at 1.8 V, 0–130 ns)	Measured

10. Conclusion

A 4-bit ADD/SUB ALU was designed in static CMOS in the gpdk045 technology using a hierarchy of reusable cells: inverter, NAND2, NOR2 and XOR2 cells, a full adder of 58 transistors, a one-bit slice that adds a conditional inverter on B, and a four-slice ripple-carry ALU of 302 transistors with a single output XOR2 for the carry-out and borrow flag. Subtraction is obtained at no additional adder cost through two's complement arithmetic. The one-bit slice passed all 16 input combinations and the ALU passed all 13 directed cases in both ADD and SUB modes, including the carry-out and borrow boundary cases, and the outputs swing fully between the rails. The design is therefore verified at the functional level.

Against the objectives of Section 4, the equations, the CMOS schematics, the symbols, the testbenches, the 13 test cases, the area estimate, the propagation delay and the average power were delivered. The worst-case carry-path delay is 275.7 ps and the average power over the 13-case run is about 10.25 µW, both at a 1.8 V supply, a 10 fF load and 27 °C. The design was simulated only at a 1.8 V supply. The recommended next steps are therefore to repeat the simulations at the nominal supply of the kit, to resize the cells with wider PMOS devices if speed matters, and, if the course requires it, to proceed to layout and post-layout simulation.

References

[1] N. H. E. Weste and D. M. Harris, CMOS VLSI Design: A Circuits and Systems Perspective, 4th ed. Addison-Wesley, 2011. ISBN 978-0-321-54774-3.

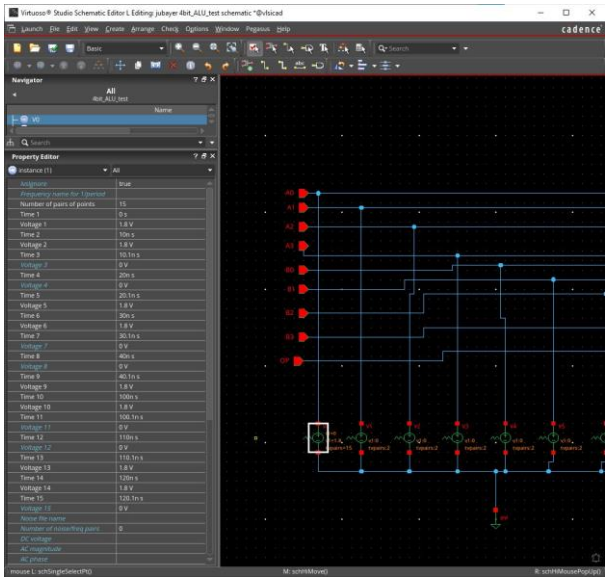
[2] J. M. Rabaey, A. Chandrakasan and B. Nikolic, Digital Integrated Circuits: A Design Perspective, 2nd ed. Pearson, 2003. ISBN 978-0-13-090996-1.

Appendix A: Stimulus Definition for the ALU Testbench

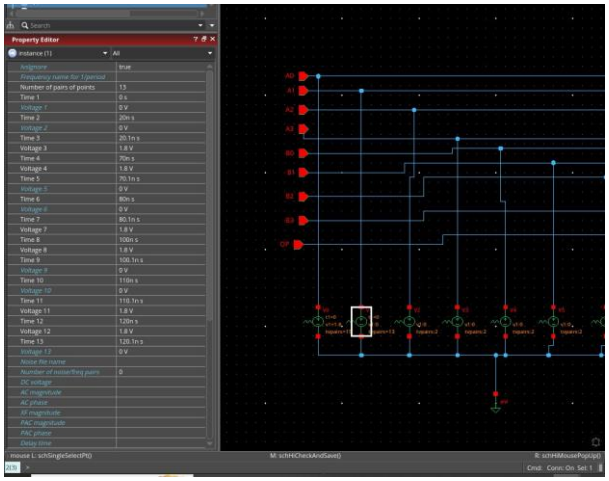
Table 19 lists the piecewise-linear sources V0 to V8 as entered in the testbench. Times are in nanoseconds; high is 1.8 V and low is 0 V. Each change is entered as two points, the old level at the change time and the new level 0.1 ns later. The property-editor screenshots that follow (Figure 28) show the values that the simulator received for each source.

Table 19. vpwl source definitions.

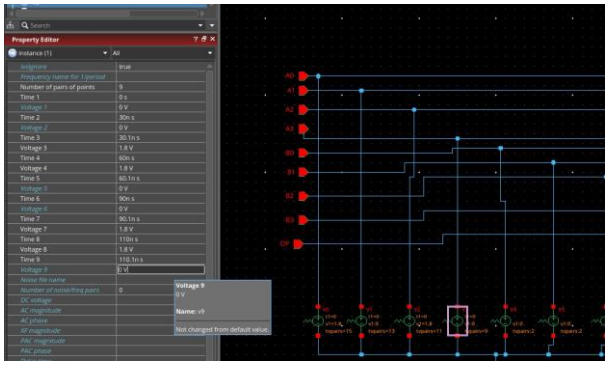
Source	Drives	Pairs	Waveform shape (ns)
V0	A0	15	high 0–10, low 10–20, high 20–30, low 30–40, high 40–100, low 100–110, high 110–120, low after 120
V1	A1	13	low 0–20, high 20–70, low 70–80, high 80–100, low 100–110, high 110–120, low after 120
V2	A2	11	high 0–10, low 10–20, high 20–30, low 30–40, high 40–100, low after 100
V3	A3	9	low 0–30, high 30–60, low 60–90, high 90–110, low after 110
V4	B0	13	high 0–10, low 10–20, high 20–60, low 60–70, high 70–90, low 90–100, high after 100
V5	B1	7	high 0–10, low 10–50, high 50–90, low after 90
V6	B2	17	low 0–30, high 30–40, low 40–50, high 50–60, low 60–80, high 80–90, low 90–110, high 110–120, low after 120
V7	B3	5	low 0–50, high 50–60, low after 60
V8	OP	3	low 0–60, high after 60



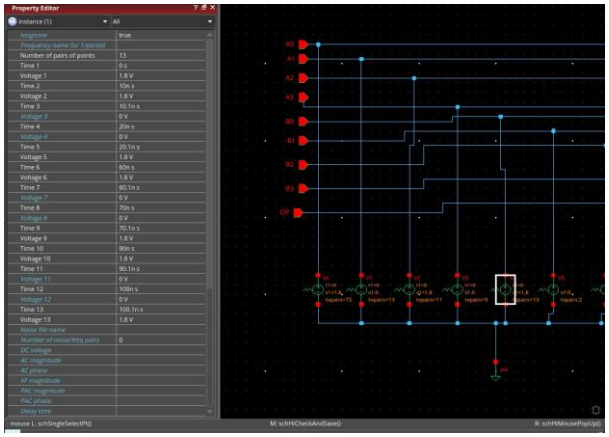
V0 = A0, 15 pairs



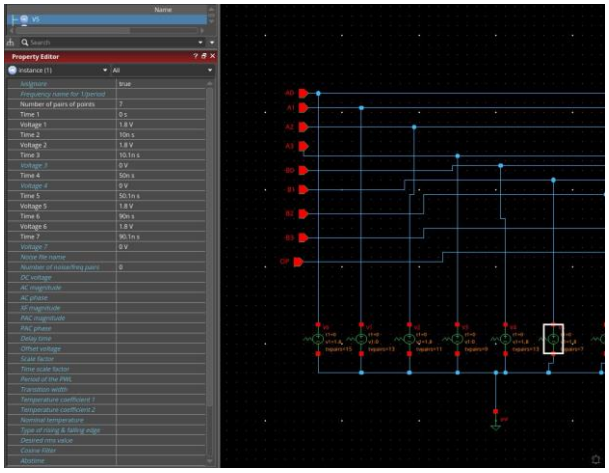
V1 = A1, 13 pairs



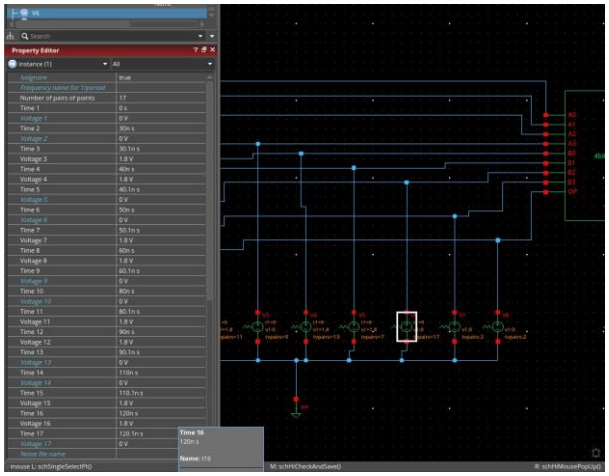
V3 = A3, 9 pairs



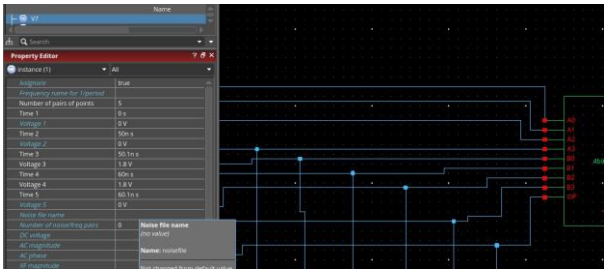
V4 = B0, 13 pairs



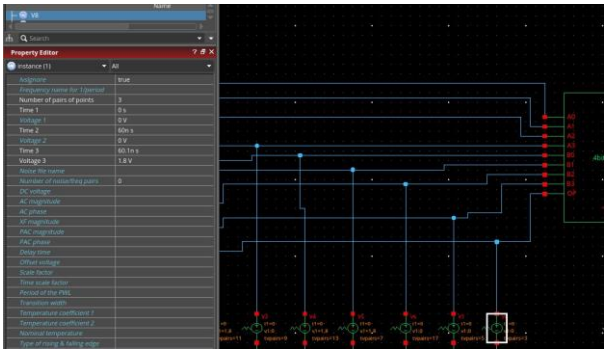
V5 = B1, 7 pairs, voltages 0 / 1.8 V (after the fix)



V6 = B2, 17 pairs



V7 = B3, 5 pairs



V8 = OP, 3 pairs: low to 60 ns, high after

Figure 28. Property-editor screenshots of the vpwl sources V0, V1, V3–V8.

The V2 (A2) source is shown only in its state before correction (Figure 31, Appendix C).

Appendix B: Deviations from the Proposal

The project proposal was submitted on 12 October 2026. The implemented design and the work reported here differ from it in the points of Table 20. The proposal values were assumptions made before the design was built; the as-built values come from the Cadence netlists and schematics.

Table 20. Deviations from the proposal.

Item	Proposal	As built	Remark
Technology	gpdk180, 180 nm	gpdk045, 45 nm	The proposal entry was an error; gpdk045 is used throughout.
Transistor sizes	L = 180 nm; Wp = 2 µm, Wn = 1–2 µm (ratioed)	W = 120 nm, L = 45 nm, all transistors, no ratioing	See Section 7.2 and 9.6.
XOR2	12 transistors	14 transistors (3 inverters + 8)	Input and output inverters.
Full adder	2 XOR2 + 3 NAND2 (NAND-NAND carry), 36 transistors	2 XOR2 + 3 NAND2 + 5 inverters + 2 NOR2, 58 transistors	Majority carry; same function.
Slice and ALU counts	48 and 204 transistors	72 and 302 transistors	Derived from the as-built cells.
Critical path	9 levels to R3, 11 to OVF	About 20 to R3, 26 to OVF	Hand count for the as-built full adder.
Pin names	A<3:0>, B<3:0>, R<3:0>, VDD, GND	A0–A3, B0–B3, R0–R3, vdd, gnd; vgs on xor, vss on nand	Not renamed; see Section 5.7.
Supply voltage	1.8 V (for a 180 nm kit)	1.8 V kept on a 45 nm kit	Justification and limitation in Section 5.4.
Corner	27 °C, typical-typical	27 °C; model section mc, Maestro corner nom	confirmed
Input transition	100 ps on all sources	100 ps (slice) and 0.1 ns (ALU); 1 ns in the full-adder and XOR2 test cells	Cell-level tests used slower edges.
Transient run	140 ns for the ALU	150 ns for the ALU, 160 ns for the slice	Covers all 13 cases (0–130 ns).
Delay and power	Planned	Measured: 275.7 ps worst-case delay; ≈ 10.25 µW average power	Section 9.4 and 9.5.
OP-switch delay test	A = B = 0000	Not run; use A = 0111, B = 0001, OP 0 → 1	A = B = 0000 gives no output transition.
Separate cell testbenches	INV, NAND2, XOR2 and full adder tested separately	Test cells exist for the XOR2 and the full adder; none for INV and NAND2; results of the cell tests are not included	Covered by the exhaustive slice test.
512-combination sweep	If time permits	Not done	Directed ALU test plus exhaustive slice test.

Appendix C: Screenshots of Errors Found and Corrected

The figures below document the errors summarised in Table 14.

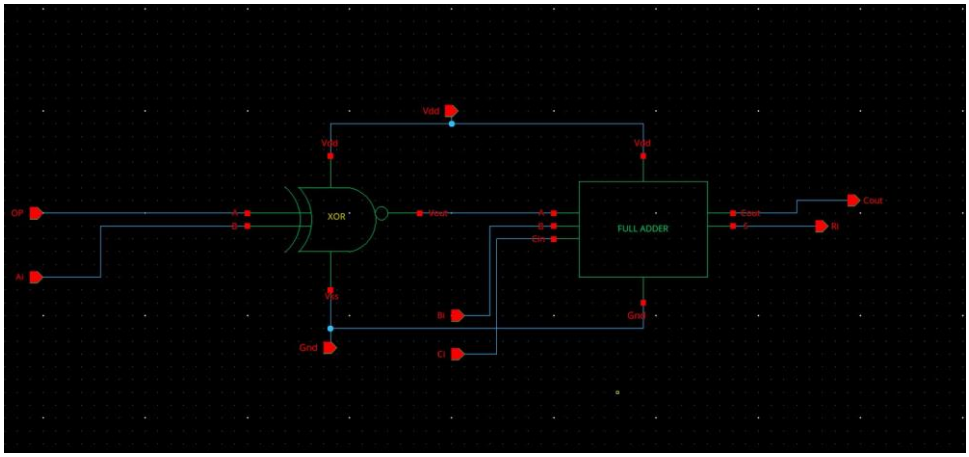
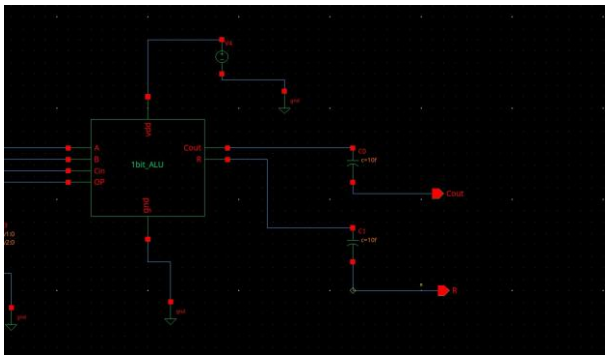
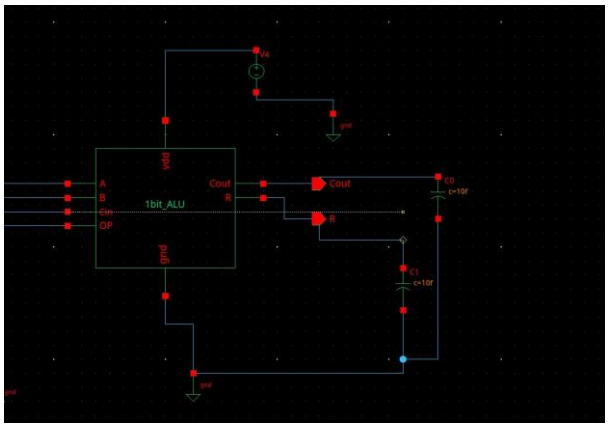


Figure 29. First attempt (incorrect): XOR on OP and A, with B bypassing the XOR.

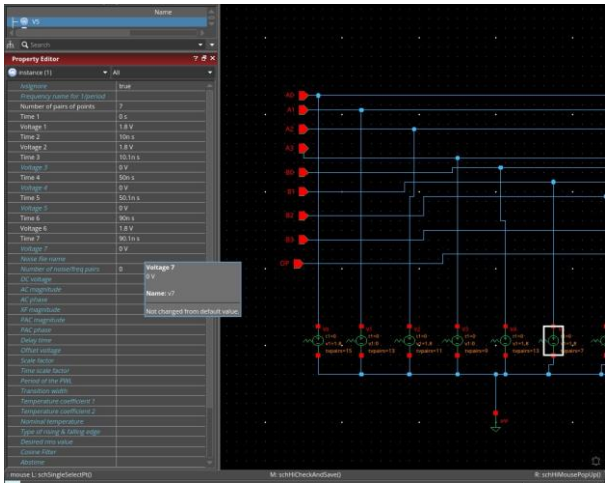


(a) Capacitors in series with the outputs

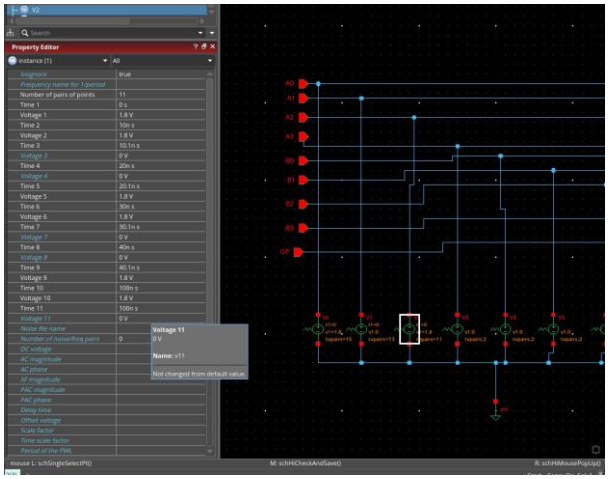


(b) Capacitor wire not joined to its net

Figure 30. Load-capacitor errors (incorrect).



(a) V5 before the fix: Voltage 5 = 0 V



(b) V2 before the fix: Time 11 = 100n, same as Time 10

Figure 31. The two vpwl entry errors before correction.